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Organometallic compound, light-emitting device including the same, and electronic apparatus including the light-emitting device

Abstract

Provided are an organometallic compound represented by Formula 1, a light-emitting device including the organometallic compound, and an electronic apparatus including the light-emitting device: ##STR00001## Each substituent in Formula 1 is the same as described in the specification.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

(1) This application claims priority to and benefits of Korean Patent Application No. 10-2021-0037436 under 35 U.S.C. § 119, filed on Mar. 23, 2021 in the Korean Intellectual Property Office, the entire contents of which are incorporated herein by reference.

BACKGROUND

1. Technical Field

(2) Embodiments relate to an organometallic compound, a light-emitting device including the same, and an electronic apparatus including the light-emitting device.

2. Description of the Related Art

(3) Organic light-emitting devices (OLEDs) are self-emissive devices that, as compared with devices of the related art, have wide viewing angles, high contrast ratios, short response times, and excellent characteristics in terms of luminance, driving voltage, and response speed, and produce full-color images.

(4) Organic light-emitting devices may include a first electrode located on a substrate, and a hole transport region, an emission layer, an electron transport region, and a second electrode sequentially stacked on the first electrode. Holes provided from the first electrode may move toward the emission layer through the hole transport region, and electrons provided from the second electrode may move toward the emission layer through the electron transport region. Carriers, such as holes and electrons, recombine in the emission layer to produce excitons. These excitons transition from an excited state to a ground state, thereby generating light.

(5) It is to be understood that this background of the technology section is, in part, intended to provide useful background for understanding the technology. However, this background of the technology section may also include ideas, concepts, or recognitions that were not part of what was known or appreciated by those skilled in the pertinent art prior to a corresponding effective filing date of the subject matter disclosed herein.

SUMMARY

(6) Provided are an organometallic compound, a light-emitting device having a low driving voltage, high luminance, high efficiency, and long lifespan characteristics due to the inclusion of the organometallic compound, and an electronic apparatus including the light-emitting device.

(7) Additional aspects will be set forth in part in the description, which follows and, in part, will be apparent from the description, or may be learned by practice of the embodiments of the disclosure.

(8) An organometallic compound according to an embodiment of the disclosure may be represented by Formula 1.

(9) ##STR00002##

(10) In Formula 1, M.sub.1 and M.sub.2 may each independently be platinum (Pt), palladium (Pd),

copper (Cu), silver (Ag), gold (Au), rhodium (Rh), iridium (Ir), ruthenium (Ru), osmium (Os), titanium (Ti), zirconium (Zr), hafnium (Hf), europium (Eu), terbium (Tb), or thulium (Tm), A1 to A4 and B1 to B4 may each independently be a C.sub.5-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, Y.sub.10 to Y.sub.12, Y.sub.20 to Y.sub.22, Y.sub.30 to Y.sub.32, Y.sub.40 to Y.sub.42, Y.sub.50 to Y.sub.52, Y.sub.60 to Y.sub.62, Y.sub.70 to Y.sub.72, and Y.sub.80 to Y.sub.82 may each independently be C or N, T.sub.1 and T.sub.2 may each independently be a single bond, *—O—*, *—S—*, *—C(R.sub.1)(R.sub.2)—*, *—C(R.sub.1)=*, *=C(R.sub.1)—*, *—C(R.sub.1)=C(R.sub.2)—*, *—C(=O)—*, *—C(=S)—*, *—C≡C—*, *—B(R.sub.1)—*, *—N(R.sub.1)—*, *—P(R.sub.1)—*, *—Si(R.sub.1)(R.sub.2)—*, or *—Ge(R.sub.1)(R.sub.2)—*, L.sub.10 may be a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, a10 may be 1, 2, 3, 4, or 5, R.sub.1, R.sub.2, R.sub.10, R.sub.20, R.sub.30, R.sub.40, R.sub.50, R.sub.60, R.sub.70, and R.sub.80 may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.20 alkoxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.10 cycloalkyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.10 heterocycloalkyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.10 cycloalkenyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.10 heterocycloalkenyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 aryl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 aryloxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 arylthio group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heteroaryl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heteroaryloxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heteroarylthio group unsubstituted or substituted with at least one R.sub.10a, a monovalent non-aromatic condensed polycyclic group unsubstituted or substituted with at least one R.sub.10a, a monovalent non-aromatic condensed heteropolycyclic group unsubstituted or substituted with at least one R.sub.10a, —Si(Q.sub.41)(Q.sub.42)(Q.sub.43), —N(Q.sub.41)(Q.sub.42), —B(Q.sub.41)(Q.sub.42), —C(=O)(Q.sub.41), —S(=O).sub.2(Q.sub.41), or —P(=O)(Q.sub.41)(Q.sub.42), two neighboring substituents among R.sub.1, R.sub.2, R.sub.10, R.sub.20, R.sub.30, R.sub.40, R.sub.50, R.sub.60, R.sub.70, and R.sub.80 may optionally be linked to each other to form a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, b10, b20, b30, b40, b50, b60, b70, and b80 may each independently be 1, 2, 3, 4, 5, 6, 7, or 8, * and *' each indicate a binding site to a neighboring atom, and R.sub.10a may be: deuterium (-D), —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, or a nitro group; a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, or a C.sub.1-C.sub.60 alkoxy group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryloxy group, a C.sub.1-C.sub.60 heteroarylthio group, —Si(Q.sub.11)(Q.sub.12)(Q.sub.13), —N(Q.sub.11)(Q.sub.12), —B(Q.sub.11)(Q.sub.12), —C(=O)(Q.sub.11), —S(=O).sub.2(Q.sub.11), —P(=O)(Q.sub.11)(Q.sub.12), or any combination thereof; a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryloxy group, or a C.sub.1-C.sub.60 heteroarylthio group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a

C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryloxy group, a C.sub.1-C.sub.60 heteroarylthio group, —Si(Q.sub.21)(Q.sub.22)(Q.sub.23), —N(Q.sub.21)(Q.sub.22), —B(Q.sub.21)(Q.sub.22), —C(=O)(Q.sub.21), —S(=O).sub.2(Q.sub.21), —P(=O)(Q.sub.21)(Q.sub.22), or any combination thereof; or —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), or —P(=O)(Q.sub.31)(Q.sub.32), wherein Q.sub.11 to Q.sub.13, Q.sub.21 to Q.sub.23, Q.sub.31 to Q.sub.33, and Q.sub.41 to Q.sub.43 may each independently be hydrogen; deuterium; —F; —Cl; —Br; —I; a hydroxyl group; a cyano group; a nitro group; a C.sub.1-C.sub.60 alkyl group; a C.sub.2-C.sub.60 alkenyl group; a C.sub.2-C.sub.60 alkynyl group; a C.sub.1-C.sub.60 alkoxy group; or a C.sub.3-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, each unsubstituted or substituted with deuterium, —F, a cyano group, a C.sub.1-C.sub.60 alkyl group, a C.sub.1-C.sub.60 alkoxy group, a phenyl group, a biphenyl group, or any combination thereof.

(11) In an embodiment, M.sub.1 and M.sub.2 may each independently be Pt or Pd.

(12) In an embodiment, A1 to A4 may each independently be a 5-membered heterocyclic group or may each independently be a polycyclic C.sub.1-C.sub.30 heterocyclic group comprising a 5-membered heterocyclic group.

(13) In an embodiment, A1 and A4 may each independently be a group represented by one of Formulae 3-1 to 3-3, which are explained below, and A2 and A3 may each independently be a group represented by one of Formulae 3-4 and 3-5, which are explained below.

(14) In an embodiment, a bond between M.sub.1 and Y.sub.10 may be a coordinate bond, a bond between M.sub.1 and Y.sub.20 may be a coordinate bond, a bond between M.sub.2 and Y.sub.30 may be a coordinate bond, a bond between M.sub.2 and Y.sub.40 may be a coordinate bond, a bond between M.sub.1 and Y.sub.50 may be a covalent bond, a bond between M.sub.1 and Y.sub.60 may be a covalent bond, a bond between M.sub.2 and Y.sub.70 may be a covalent bond, and a bond between M.sub.2 and Y.sub.80 may be a covalent bond.

(15) In an embodiment, T.sub.1 and T.sub.2 may each independently be *—O—*' or *—S—*'.

(16) In an embodiment, L.sub.10 may be a C.sub.1-C.sub.20 alkylene group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.20 alkenylene group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.20 alkynylene group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.10 cycloalkylene group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.10 heterocycloalkylene group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.10 cycloalkenylene group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.10 heterocycloalkenylene group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 arylene group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heteroarylene group unsubstituted or substituted with at least one R.sub.10a, a divalent non-aromatic condensed polycyclic group unsubstituted or substituted with at least one R.sub.10a, or a divalent non-aromatic condensed heteropolycyclic group unsubstituted or substituted with at least one R.sub.10a.

(17) In an embodiment, L.sub.10 may be a group represented by one of Formulae 4-1 to 4-3, which are explained below.

(18) In an embodiment, R.sub.1, R.sub.2, R.sub.10, R.sub.20, R.sub.30, R.sub.40, R.sub.50, R.sub.60, R.sub.70, and R.sub.80 may each independently be: hydrogen, deuterium, —F, —Cl, —Br, —I, a cyano group, a C.sub.1-C.sub.20 alkyl group, or a C.sub.1-C.sub.20 alkoxy group; or a C.sub.1-C.sub.20 alkyl group or a C.sub.1-C.sub.20 alkoxy group, each substituted with deuterium, —F, —Cl, —Br, —I, a cyano group, a phenyl group, a biphenyl group, or any combination thereof; or a group represented by one of Formulae 5-1 to 5-26 and 6-1 to 6-55, which are explained below. Neighboring two or more of R.sub.1, R.sub.2, R.sub.10, R.sub.20, R.sub.30, R.sub.40, R.sub.50, R.sub.60, R.sub.70, and R.sub.80 may be optionally linked to each other to form a cyclopentane

group, a cyclohexane group, a benzene group, a naphthylene group, a fluorene group, a pyridine group, a pyrimidine group, a triazine group, or a carbazole group.

(19) In an embodiment, the organometallic compound may be represented by one of Formulae 11-1 to 11-3, which are explained below.

(20) In an embodiment, the organometallic compound may be electrically neutral.

(21) In an embodiment, the organometallic compound may be one selected from Compounds BD1 to BD104, which are explained below.

(22) Another aspect of the disclosure provides a light-emitting device which may include a first electrode, a second electrode facing the first electrode, an interlayer disposed between the first electrode and the second electrode and including an emission layer, and the organometallic compound.

(23) In an embodiment, the first electrode may be an anode, the second electrode may be a cathode, and the interlayer may further include a hole transport region disposed between the first electrode and the emission layer and an electron transport region disposed between the emission layer and the second electrode. The hole transport region may include a hole injection layer, a hole transport layer, an emission auxiliary layer, an electron blocking layer, or any combination thereof, and the electron transport region may include a hole blocking layer, an electron transport layer, an electron injection layer, or any combination thereof.

(24) In an embodiment, the emission layer may include the organometallic compound.

(25) In an embodiment, the emission layer may include a host and a dopant, and the dopant may include the organometallic compound.

(26) In an embodiment, the hole transport region may further include a p-dopant having a lowest unoccupied molecular orbital (LUMO) energy level equal to or less than about -3.5 eV.

(27) In an embodiment, the electron transport region may include the electron transport layer and the electron injection layer, and at least one layer of the electron transport layer and the electron injection layer may include an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal compound, an alkaline earth-metal compound, a rare earth metal compound, an alkali metal complex, an alkaline earth-metal complex, a rare earth metal complex, or any combination thereof.

(28) Another aspect of the disclosure provides an electronic apparatus which may include the light-emitting device.

(29) In an embodiment, the electronic apparatus may further include a thin-film transistor. The thin-film transistor may include a source electrode and a drain electrode, and the first electrode of the light-emitting device may be electrically connected to the source electrode or the drain electrode.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other aspects and features of the disclosure will become more apparent by describing in detail embodiments thereof with reference to the accompanying drawings, in which:

(2) FIG. 1 is a schematic cross-sectional view of a light-emitting device according to an embodiment;

(3) FIG. 2 is a schematic cross-sectional view of an electronic apparatus according to an embodiment; and

(4) FIG. 3 is a schematic cross-sectional view of an electronic apparatus according to an embodiment.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(5) The disclosure will now be described more fully hereinafter with reference to the accompanying drawings, in which embodiments are shown. This disclosure may, however, be embodied in different forms and should not be construed as limited to the embodiments set forth herein. Rather,

these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the disclosure to those skilled in the art.

(6) In the drawings, the sizes, thicknesses, ratios, and dimensions of the elements may be exaggerated for ease of description and for clarity. Like numbers refer to like elements throughout.

(7) In the description, it will be understood that when an element (or region, layer, part, etc.) is referred to as being “on”, “connected to”, or “coupled to” another element, it can be directly on, connected to, or coupled to the other element, or one or more intervening elements may be present therebetween. In a similar sense, when an element (or region, layer, part, etc.) is described as “covering” another element, it can directly cover the other element, or one or more intervening elements may be present therebetween.

(8) In the description, when an element is “directly on,” “directly connected to,” or “directly coupled to” another element, there are no intervening elements present. For example, “directly on” may mean that two layers or two elements are disposed without an additional element such as an adhesion element therebetween.

(9) As used herein, the expressions used in the singular such as “a,” “an,” and “the,” are intended to include the plural forms as well, unless the context clearly indicates otherwise.

(10) As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. For example, “A and/or B” may be understood to mean “A, B, or A and B.” The terms “and” and “or” may be used in the conjunctive or disjunctive sense and may be understood to be equivalent to “and/or”.

(11) In the specification and the claims, the term “at least one of” is intended to include the meaning of “at least one selected from the group of” for the purpose of its meaning and interpretation. For example, “at least one of A and B” may be understood to mean “A, B, or A and B.” When preceding a list of elements, the term, “at least one of,” modifies the entire list of elements and does not modify the individual elements of the list.

(12) When an embodiment is implemented differently, a specific process order may be performed differently from the described order. For example, two processes described in succession may be performed substantially simultaneously, or may be performed in an order opposite to the described order.

(13) It will be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another element. Thus, a first element could be termed a second element without departing from the teachings of the disclosure. Similarly, a second element could be termed a first element, without departing from the scope of the disclosure.

(14) The term “interlayer” as used herein refers to a single layer and/or all layers between a first electrode and a second electrode of a light-emitting device.

(15) The spatially relative terms “below”, “beneath”, “lower”, “above”, “upper”, or the like, may be used herein for ease of description to describe the relations between one element or component and another element or component as illustrated in the drawings. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation, in addition to the orientation depicted in the drawings. For example, in the case where a device illustrated in the drawing is turned over, the device positioned “below” or “beneath” another device may be placed “above” another device. Accordingly, the illustrative term “below” may include both the lower and upper positions. The device may also be oriented in other directions and thus the spatially relative terms may be interpreted differently depending on the orientations.

(16) The terms “about” or “approximately” as used herein is inclusive of the stated value and means within an acceptable range of deviation for the recited value as determined by one of ordinary skill in the art, considering the measurement in question and the error associated with measurement of the recited quantity (i.e., the limitations of the measurement system). For example, “about” may mean within one or more standard deviations, or within +20%, 10%, or $\pm 5\%$ of the

stated value.

(17) It should be understood that the terms “comprises,” “comprising,” “includes,” “including,” “have,” “having,” “contains,” “containing,” and the like are intended to specify the presence of stated features, integers, steps, operations, elements, components, or combinations thereof in the disclosure, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, or combinations thereof.

(18) Unless otherwise defined or implied herein, all terms (including technical and scientific terms) used have the same meaning as commonly understood by those skilled in the art to which this disclosure pertains. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and should not be interpreted in an ideal or excessively formal sense unless clearly defined in the specification.

(19) According to an aspect of the disclosure, an organometallic compound may be represented by Formula 1:

(20) ##STR00003##

(21) In Formula 1, M.sub.1 and M.sub.2 may each independently be platinum (Pt), palladium (Pd), copper (Cu), silver (Ag), gold (Au), rhodium (Rh), iridium (Ir), ruthenium (Ru), osmium (Os), titanium (Ti), zirconium (Zr), hafnium (Hf), europium (Eu), terbium (Tb), or thulium (Tm).

(22) In an embodiment, M.sub.1 and M.sub.2 may each independently be Pt or Pd.

(23) In Formula 1, A1 to A4 and B1 to B4 may each independently be a C.sub.5-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group.

(24) In an embodiment, A1 to A4 may each independently be a 5-membered heterocyclic group or may each independently be a polycyclic C.sub.1-C.sub.30 heterocyclic group including a 5-membered heterocyclic group.

(25) In an embodiment, A1 to A4 and B1 to B4 may each independently be a benzene group, a naphthalene group, an anthracene group, a phenanthrene group, an azulene group, a triphenylene group, a pyrene group, a chrysene group, a cyclopentadiene group, a 1,2,3,4-tetrahydronaphthalene group, a furan group, a thiophene group, a silole group, an indene group, a fluorene group, an indole group, a carbazole group, a benzofuran group, a dibenzofuran group, a benzothiophene group, a dibenzothiophene group, a benzosilole group, a dibenzosilole group, an indenopyridine group, an indolopyridine group, a benzofuropyridine group, a benzothienopyridine group, a benzosilolopyridine group, an indenopyrimidine group, an indolopyrimidine group, a benzofuopyrimidine group, a benzothienopyrimidine group, a benzosilolopyrimidine group, a dihydropyridine group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a quinoline group, an isoquinoline group, a quinoxaline group, a quinazoline group, a phenanthroline group, a pyrrole group, a pyrazole group, an imidazole group, a 2,3-dihydroimidazole group, a triazole group, a 2,3-dihydrotriazole group, an oxazole group, an isoxazole group, a thiazole group, an isothiazole group, an oxadiazole group, a thiadiazole group, a benzopyrazole group, a pyrazolopyridine group, a furopyrazole group, a thienopyrazole group, a benzimidazole group, a 2,3-dihydrobenzimidazole group, an imidazopyridine group, a 2,3-dihydroimidazopyridine group, a furoimidazole group, a thienimidazole group, an imidazopyrimidine group, a 2,3-dihydroimidazopyrimidine group, an imidazopyrazine group, a 2,3-dihydroimidazopyrazine group, a benzoxazole group, a benzothiazole group, a benzoxadiazole group, a benzothiadiazole group, a 5,6,7,8-tetrahydroisoquinoline group, or a 5,6,7,8-tetrahydroquinoline group.

(26) In an embodiment, A1 to A4 may each independently be a pyrrole group, a pyrazole group, an imidazole group, a 2,3-dihydroimidazole group, a triazole group, a 2,3-dihydrotriazole group, an oxazole group, an isoxazole group, a thiazole group, an isothiazole group, an oxadiazole group, a thiadiazole group, a benzopyrazole group, a pyrazolopyridine group, a furopyrazole group, a thienopyrazole group, a benzimidazole group, a 2,3-dihydrobenzimidazole group, an

imidazopyridine group, a 2,3-dihydroimidazopyridine group, a furoimidazole group, a thienimidazole group, an imidazopyrimidine group, a 2,3-dihydroimidazopyrimidine group, an imidazopyrazine group, a 2,3-dihydroimidazopyrazine group, a benzoxazole group, a benzothiazole group, a benzoxadiazole group, or a benzothiadiazole group.

(27) In an embodiment, B1 to B4 may each independently be a benzene group, a naphthalene group, a phenanthrene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, an imidazole group, a 2,3-dihydroimidazole group, an imidazopyrimidine group, a benzimidazole group, an indene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, an indole group, a pyridine group, a pyrimidine group, a carbazole group, a benzocarbazole group, a dibenzocarbazole group, an azacarbazole group, a furan group, a benzofuran group, a dibenzofuran group, an azadibenzofuran group, a naphthofuran group, a benzonaphthofuran group, a dinaphthofuran group, a thiophene group, a benzothiophene group, a dibenzothiophene group, an azadibenzothiophene group, a naphthothiophene group, a benzonaphthothiophene group, or a dinaphthothiophene group.

(28) In an embodiment, A1 to A4 and B1 to B4 may each independently be a group represented by one of Formulae 2-1 to 2-43:

(29) ##STR00004## ##STR00005## ##STR00006## ##STR00007## ##STR00008##

(30) In Formulae 2-1 to 2-43, X.sub.21 to X.sub.23 may each independently be C(Z.sub.24) or C—*, wherein at least two of X.sub.21 to X.sub.23 may be C—*, X.sub.24 may be N—*, and X.sub.25 and X.sub.26 may each independently be C(Z.sub.24) or C—*, wherein at least one of X.sub.25 and X.sub.26 may be C—*, X.sub.27 and X.sub.28 may each independently be N, N(Z.sub.25), or N—*, and X.sub.29 may be C(Z.sub.24) or C—*. At least one of X.sub.27 and X.sub.28 may be N—*, and X.sub.29 may be C—*; or X.sub.27 and X.sub.28 may each be N—*, and X.sub.29 may be C(Z.sub.24), Z.sub.21 to Z.sub.25 may each independently be the same as described in connection with R.sub.10, c21 may be 1, 2, or 3, c22 may be 1, 2, 3, 4, or 5, c23 may be 1, 2, 3, or 4, c24 may be 1 or 2, and * indicates a binding site to a neighboring atom.

(31) In an embodiment, Z.sub.21 to Z.sub.25 in Formulae 2-1 to 2-43 may each independently be: hydrogen, deuterium, —F, —Cl, —Br, —I, a cyano group, a C.sub.1-C.sub.20 alkyl group, or a C.sub.1-C.sub.20 alkoxy group; a C.sub.1-C.sub.20 alkyl group or a C.sub.1-C.sub.20 alkoxy group, each substituted with deuterium, —F, —Cl, —Br, —I, a cyano group, a phenyl group, a biphenyl group, or any combination thereof; a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a carbazolyl group, an acridinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, or a dibenzocarbazolyl group; a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a carbazolyl group, an acridinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, or a dibenzocarbazolyl group, each substituted with deuterium, —F, —Cl, —Br, —I, a cyano group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, or any combination thereof; or —Si(Q.sub.41)(Q.sub.42)(Q.sub.43) or —N(Q.sub.41)(Q.sub.42). Q.sub.41, Q.sub.42, and Q.sub.43 may each independently be the same as described in the Specification.

(32) In an embodiment, A1 to A4 may each independently be a group represented by one of Formulae 2-10 to 2-43.

(33) In an embodiment, A1 and A4 may each independently be a group represented by one of Formulae 3-1 to 3-3, and A2 and A3 may each independently be a group represented by one of Formulae 3-4 and 3-5:

(34) ##STR00009##

- (35) In Formulae 3-1 to 3-5, X.sub.31 to X.sub.39 may each independently be C(Z.sub.32) or N, Z.sub.31 and Z.sub.32 may each independently be the same as described in connection with R.sub.10, and *, *', and *'' each indicate a binding site to a neighboring atom.
- (36) In an embodiment, Z.sub.31 and Z.sub.32 in Formulae 3-1 to 3-5 may each independently be: hydrogen, deuterium, —F, —Cl, —Br, —I, a cyano group, a C.sub.1-C.sub.20 alkyl group, or a C.sub.1-C.sub.20 alkoxy group, each substituted with deuterium, —F, —Cl, —Br, —I, a cyano group, a phenyl group, a biphenyl group, or any combination thereof; a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a carbazolyl group, an acridinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, or a dibenzocarbazolyl group; a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a carbazolyl group, an acridinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, or a dibenzocarbazolyl group, each substituted with deuterium, —F, —Cl, —Br, —I, a cyano group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, or any combination thereof; or —Si(Q.sub.41)(Q.sub.42)(Q.sub.43) or —N(Q.sub.41)(Q.sub.42), wherein Q.sub.41, Q.sub.42, and Q.sub.43 may each independently be the same as described in the Specification, and *, *', and *'' each indicate a binding site to a neighboring atom.
- (37) In Formula 1, Y.sub.10 to Y.sub.12, Y.sub.20 to Y.sub.22, Y.sub.30 to Y.sub.32, Y.sub.40 to Y.sub.42, Y.sub.50 to Y.sub.52, Y.sub.60 to Y.sub.62, Y.sub.70 to Y.sub.72, and Y.sub.80 to Y.sub.82 may each independently be C or N.
- (38) In an embodiment, Y.sub.50, Y.sub.60, Y.sub.70, and Y.sub.80 may each be C.
- (39) In an embodiment, Y.sub.10 and Y.sub.20 may each be C.
- (40) In an embodiment, a bond between M.sub.1 and Y.sub.10 may be a coordinate bond, a bond between M.sub.1 and Y.sub.20 may be a coordinate bond, a bond between M.sub.2 and Y.sub.30 may be a coordinate bond, a bond between M.sub.2 and Y.sub.40 may be a coordinate bond, a bond between M.sub.1 and Y.sub.50 may be a covalent bond, a bond between M.sub.1 and Y.sub.60 may be a covalent bond, a bond between M.sub.2 and Y.sub.70 may be a covalent bond, and a bond between M.sub.2 and Y.sub.80 may be a covalent bond.
- (41) In Formula 1, T.sub.1 and T.sub.2 may each independently be a single bond, *—O—*', *—S—*', *—C(R.sub.1)(R.sub.2)—*', *—C(R.sub.1)=*', *=C(R.sub.1)—*', *—C(R.sub.1)=C(R.sub.2)—*', *—C(=O)—*', *—C(=S)—*', *—C≡C—*', *—B(R.sub.1)—*', *—N(R.sub.1)—*', *—P(R.sub.1)—*', *—Si(R.sub.1)(R.sub.2)—*', or *—Ge(R.sub.1)(R.sub.2)—*.
- (42) In an embodiment, T.sub.1 and T.sub.2 may each independently be *—O—*' or *—S—*.
- (43) In Formula 1, L.sub.10 may be a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a.
- (44) In an embodiment, L.sub.10 may be a C.sub.1-C.sub.20 alkylene group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.20 alkenylene group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.20 alkynylene group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.10 cycloalkylene group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.10 heterocycloalkylene group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.10 cycloalkenylene group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.10 heterocycloalkenylene group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 arylene group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heteroarylene group unsubstituted or substituted with at least one R.sub.10a, a divalent non-aromatic condensed

polycyclic group unsubstituted or substituted with at least one R.sub.10a, or a divalent non-aromatic condensed heteropolycyclic group unsubstituted or substituted with at least one R.sub.10a.

(45) In an embodiment, L.sub.10 may be: a phenylene group, a pentalenylene group, an indenylene group, a naphthylene group, an azulenylene group, a heptalenylene group, an indacenylene group, an acenaphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a spiro-fluorene-benzofluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenalenylene group, a phenanthrenylene group, an anthracenylene group, a fluoranthenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a naphthacenylene group, a picenylene group, a perylenylene group, a pentaphenylene group, a hexacenylene group, a pentacenylene group, a rubicenylene group, a coronenylene group, an ovalenylene group, a pyrrolylene group, a thiophenylene group, a furanylene group, an imidazolylene group, a pyrazolylene group, a thiazolylene group, an isothiazolylene group, an oxazolylene group, an isoxazolylene group, a pyridinylene group, a pyrazinylene group, a pyrimidinylene group, a pyridazinylene group, an isoindolylene group, an indolylene group, an indazolylene group, a purinylene group, a quinolinylene group, an isoquinolinylene group, a benzoquinolinylene group, a phthalazinylene group, a naphthyridinylene group, a quinoxalinylene group, a quinazolinylene group, a cinnolinylene group, a carbazolylene group, a phenanthridinylene group, an acridinylene group, a phenanthrolinylene group, a phenazinylene group, a benzimidazolylene group, a benzofuranylene group, a benzothiophenylene group, an isobenzothiazolylene group, a benzoxazolylene group, an isobenzoxazolylene group, a triazolylene group, a tetrazolylene group, an oxadiazolylene group, a triazinylene group, a dibenzofuranylene group, a dibenzothiophenylene group, a dibenzosilolylene group, a benzocarbazolylene group, a dibenzocarbazolylene group, a thiadiazolylene group, an imidazopyridinylene group, or an imidazopyrimidinylene group; or a phenylene group, a pentalenylene group, an indenylene group, a naphthylene group, an azulenylene group, a heptalenylene group, an indacenylene group, an acenaphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a spiro-fluorene-benzofluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenalenylene group, a phenanthrenylene group, an anthracenylene group, a fluoranthenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a naphthacenylene group, a picenylene group, a perylenylene group, a pentaphenylene group, a hexacenylene group, a pentacenylene group, a rubicenylene group, a coronenylene group, an ovalenylene group, a pyrrolylene group, a thiophenylene group, a furanylene group, an imidazolylene group, a pyrazolylene group, a thiazolylene group, an isothiazolylene group, an oxazolylene group, an isoxazolylene group, a pyridinylene group, a pyrazinylene group, a pyrimidinylene group, a pyridazinylene group, an isoindolylene group, an indolylene group, an indazolylene group, a purinylene group, a quinolinylene group, an isoquinolinylene group, a benzoquinolinylene group, a phthalazinylene group, a naphthyridinylene group, a quinoxalinylene group, a quinazolinylene group, a cinnolinylene group, a carbazolylene group, a phenanthridinylene group, an acridinylene group, a phenanthrolinylene group, a phenazinylene group, a benzimidazolylene group, a benzofuranylene group, a benzothiophenylene group, an isobenzothiazolylene group, a benzoxazolylene group, an isobenzoxazolylene group, a triazolylene group, a tetrazolylene group, an oxadiazolylene group, a triazinylene group, a dibenzofuranylene group, a dibenzothiophenylene group, a dibenzosilolylene group, a benzocarbazolylene group, a dibenzocarbazolylene group, a thiadiazolylene group, an imidazopyridinylene group, or an imidazopyrimidinylene group, each substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an

acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a dibenzosilolyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a thiadiazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), —P(=O)(Q.sub.31)(Q.sub.32), or any combination thereof.

(46) In an embodiment, Q.sub.31 to Q.sub.33 may each independently be a C.sub.1-C.sub.10 alkyl group, a C.sub.1-C.sub.10 alkoxy group, a phenyl group, a phenyl group substituted with a C.sub.1-C.sub.10 alkyl group, a biphenyl group, a terphenyl group, a naphthyl group, a pyridinyl group, a pyrimidyl group, a triazinyl group, a quinolinyl group, or an isoquinolinyl group.

(47) In an embodiment, L.sub.10 may be a group represented by one of Formulae 4-1 to 4-3:

(48) ##STR00010## ##STR00011##

(49) In Formulae 4-1 to 4-3, Y.sub.91 may be C(R.sub.91) or N, Y.sub.92 may be C(R.sub.92) or N, Y.sub.93 may be C(R.sub.93) or N, and Y.sub.94 may be C(R.sub.94) or N, R.sub.91 to R.sub.94 may each independently be the same as described in connection with R.sub.10a, and * and *' each indicate a binding site to a neighboring atom. R.sub.10a may be the same as described in the Specification.

(50) In Formula 1, a₁₀ may be 1, 2, 3, 4, or 5.

(51) In an embodiment, a₁₀ may be 1 or 2.

(52) In an embodiment, a₁₀ may be 1.

(53) In Formula 1, R.sub.1, R.sub.2, R.sub.10, R.sub.20, R.sub.30, R.sub.40, R.sub.50, R.sub.60, R.sub.70, and R.sub.80 may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.20 alkoxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.10 cycloalkyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.10 heterocycloalkyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.10 cycloalkenyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.10 heterocycloalkenyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 aryl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 aryloxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 arylthio group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heteroaryl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heteroaryloxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heteroarylthio group unsubstituted or substituted with at least one R.sub.10a, a monovalent non-aromatic condensed polycyclic group unsubstituted or substituted with at least one R.sub.10a, a monovalent non-aromatic condensed heteropolycyclic group unsubstituted or substituted with at least one R.sub.10a, —Si(Q.sub.41)(Q.sub.42)(Q.sub.43), —N(Q.sub.41)(Q.sub.42), —

B(Q.sub.41)(Q.sub.42), —C(=O)(Q.sub.41), —S(=O).sub.2(Q.sub.41), or —P(=O)(Q.sub.41) (Q.sub.42), and two neighboring substituents among R.sub.1, R.sub.2, R.sub.10, R.sub.20, R.sub.30, R.sub.40, R.sub.50, R.sub.60, R.sub.70, and R.sub.80 may optionally be linked to each other to form a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a.

(54) In an embodiment, R.sub.1, R.sub.2, R.sub.10, R.sub.20, R.sub.30, R.sub.40, R.sub.50, R.sub.60, R.sub.70, and R.sub.80 may each independently be: hydrogen, deuterium, —F —Cl, —Br, —I, a hydroxyl group, or a cyano group; a C.sub.1-C.sub.20 alkyl group, a C.sub.2-C.sub.20 alkenyl group, a C.sub.2-C.sub.20 alkynyl group, or a C.sub.1-C.sub.20 alkoxy group; a C.sub.1-C.sub.20 alkyl group, a C.sub.2-C.sub.20 alkenyl group, a C.sub.2-C.sub.20 alkynyl group, or a C.sub.1-C.sub.20 alkoxy group, each substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a naphthyl group, a pyridinyl group, a pyrimidinyl group, —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O) (Q.sub.31), —S(=O).sub.2(Q.sub.31), —P(=O)(Q.sub.31)(Q.sub.32), or any combination thereof; a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a pentalenyl group, an indenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a furanyl group, a thiophenyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzothiazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, or an imidazopyrimidinyl group; or a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a pentalenyl group, an indenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a furanyl group, a thiophenyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzothiazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a

dibenzocarbazolyl group, an imidazopyridinyl group, or an imidazopyrimidinyl group, each substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.2-C.sub.20 alkenyl group, a C.sub.2-C.sub.20 alkynyl group, a C.sub.1-C.sub.20 alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a pentalenyl group, an indenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a furanyl group, a thiophenyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzothiazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), —P(=O)(Q.sub.31)(Q.sub.32), or any combination thereof.

(55) In an embodiment, Q.sub.31 to Q.sub.33 may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a cyano group, a C.sub.1-C.sub.20 alkyl group, a C.sub.2-C.sub.20 alkenyl group, a C.sub.2-C.sub.20 alkynyl group, a C.sub.1-C.sub.20 alkoxy group, a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.20 aryl group, a C.sub.1-C.sub.20 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, a biphenyl group, or a terphenyl group.

(56) In an embodiment, R.sub.1, R.sub.2, R.sub.10, R.sub.20, R.sub.30, R.sub.40, R.sub.50, R.sub.60, R.sub.70, and R.sub.80 may each independently be: hydrogen, deuterium, —F, —Cl, —Br, —I, a cyano group, a C.sub.1-C.sub.20 alkyl group, or a C.sub.1-C.sub.20 alkoxy group; a C.sub.1-C.sub.20 alkyl group or a C.sub.1-C.sub.20 alkoxy group, each substituted with deuterium, —F, —Cl, —Br, —I, a cyano group, a phenyl group, a biphenyl group, or any combination thereof; or a group represented by one of Formulae 5-1 to 5-26 and 6-1 to 6-55:

(57) ##STR00012## ##STR00013## ##STR00014## ##STR00015## ##STR00016##
##STR00017## ##STR00018## ##STR00019## ##STR00020## ##STR00021##

(58) In Formulae 5-1 to 5-26 and 6-1 to 6-55, X.sub.51 and X.sub.52 may each independently be O, S, C(Z.sub.53)(Z.sub.54), N(Z.sub.53), or Si(Z.sub.53)(Z.sub.54), Z.sub.51 to Z.sub.54 may each independently be the same as described in connection with R.sub.10a, e2 may be 1 or 2, e3 may be 1, 2, or 3, e4 may be 1, 2, 3, or 4, e5 may be 1, 2, 3, 4, or 5, e6 may be 1, 2, 3, 4, 5, or 6, e7 may be 1, 2, 3, 4, 5, 6, or 7, e9 may be 1, 2, 3, 4, 5, 6, 7, 8, or 9, and * indicates a binding site to a neighboring atom.

(59) In an embodiment, Z.sub.51 to Z.sub.54 in Formulae 5-1 to 5-26 and 6-1 to 6-55 may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.2-C.sub.20 alkenyl group, a C.sub.2-C.sub.20 alkynyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a phenanthrenyl group, an anthracenyl group, a triphenylenyl

- group, a pyridinyl group, a pyrimidinyl group, a carbazolyl group, or a triazinyl group.
- (60) In an embodiment, two or more neighboring substituents among R.sub.1, R.sub.2, R.sub.10, R.sub.20, R.sub.30, R.sub.40, R.sub.50, R.sub.60, R.sub.70, and R.sub.80 may optionally be linked to each other to form a cyclopentane group, a cyclohexane group, a benzene group, a naphthylene group, a fluorene group, a pyridine group, a pyrimidine group, a triazine group, or a carbazole group.
- (61) In Formula 1, b10, b20, b30, b40, b50, b60, b70, and b80 may each independently be 1, 2, 3, 4, 5, 6, 7, or 8.
- (62) In an embodiment, b10, b20, b30, b40, b50, b60, b70, and b80 may each independently be 1, 2, 3, or 4.
- (63) In an embodiment, b10, b20, b30, b40, b50, b60, b70, and b80 may each independently be 1 or 2.
- (64) In Formula 1, * and *' each indicate a binding site to a neighboring atom.
- (65) R.sub.10a may be: deuterium (-D), —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, or a nitro group; a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, or a C.sub.1-C.sub.60 alkoxy group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryloxy group, a C.sub.1-C.sub.60 heteroarylthio group, —Si(Q.sub.11)(Q.sub.12)(Q.sub.13), —N(Q.sub.11)(Q.sub.12), —B(Q.sub.11)(Q.sub.12), —C(=O)(Q.sub.11), —S(=O).sub.2(Q.sub.11), —P(=O)(Q.sub.11)(Q.sub.12), or any combination thereof; a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryloxy group, or a C.sub.1-C.sub.60 heteroarylthio group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryloxy group, a C.sub.1-C.sub.60 heteroarylthio group, —Si(Q.sub.21)(Q.sub.22)(Q.sub.23), —N(Q.sub.21)(Q.sub.22), —B(Q.sub.21)(Q.sub.22), —C(=O)(Q.sub.21), —S(=O).sub.2(Q.sub.21), —P(=O)(Q.sub.21)(Q.sub.22), or any combination thereof; or —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), or —P(=O)(Q.sub.31)(Q.sub.32), wherein Q.sub.11 to Q.sub.13, Q.sub.21 to Q.sub.23, Q.sub.31 to Q.sub.33, and Q.sub.41 to Q.sub.43 may each independently be: hydrogen; deuterium; —F; —Cl; —Br; —I; a hydroxyl group; a cyano group; a nitro group; a C.sub.1-C.sub.60 alkyl group; a C.sub.2-C.sub.60 alkenyl group; a C.sub.2-C.sub.60 alkynyl group; a C.sub.1-C.sub.60 alkoxy group; or a C.sub.3-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, each unsubstituted or substituted with deuterium, —F, a cyano group, a C.sub.1-C.sub.60 alkyl group, a C.sub.1-C.sub.60 alkoxy group, a phenyl group, a biphenyl group, or any combination thereof.
- (66) In an embodiment, the organometallic compound may be represented by one of Formulae 11-1 to 11-3:
- (67) ##STR00022## ##STR00023## ##STR00024##
- (68) In Formulae 11-1 to 11-3, Y.sub.23 may be C(R.sub.23) or N, Y.sub.24 may be C(R.sub.24) or N, Y.sub.25 may be C(R.sub.25) or N, Y.sub.26 may be C(R.sub.26) or N, Y.sub.27 may be C(R.sub.27) or N, Y.sub.28 may be C(R.sub.28) or N, Y.sub.33 may be C(R.sub.33) or N, Y.sub.34 may be C(R.sub.34) or N, Y.sub.35 may be C(R.sub.35) or N, Y.sub.36 may be C(R.sub.36) or N, Y.sub.37 may be C(R.sub.37) or N, and Y.sub.38 may be C(R.sub.38) or N, R.sub.23 to R.sub.28 may each independently be the same as described in connection with R.sub.20 in Formula 1, and R.sub.33 to R.sub.38 may each independently be the same as described in connection with

R.sub.30 in Formula 1, and Y.sub.10 to Y.sub.12, Y.sub.40 to Y.sub.42, Y.sub.50 to Y.sub.52, Y.sub.60 to Y.sub.62, Y.sub.70 to Y.sub.72, Y.sub.80 to Y.sub.82, R.sub.10, R.sub.40, R.sub.50, R.sub.60, R.sub.70, R.sub.80, A1, A4, B1 to B4, M.sub.1, M.sub.2, T.sub.1, T.sub.2, b10, b40, b50, b60, b70, b80, a10, and L.sub.10 may each independently be the same as described in the specification.

(69) In an embodiment, the organometallic compound may be one selected from Compounds BD1 to BD104:

(70) ##STR00025## ##STR00026## ##STR00027## ##STR00028## ##STR00029##
##STR00030## ##STR00031## ##STR00032## ##STR00033## ##STR00034## ##STR00035##
##STR00036## ##STR00037## ##STR00038## ##STR00039## ##STR00040## ##STR00041##
##STR00042## ##STR00043## ##STR00044## ##STR00045## ##STR00046## ##STR00047##
##STR00048## ##STR00049## ##STR00050## ##STR00051## ##STR00052## ##STR00053##
##STR00054## ##STR00055## ##STR00056## ##STR00057## ##STR00058## ##STR00059##
##STR00060## ##STR00061## ##STR00062## ##STR00063## ##STR00064## ##STR00065##
##STR00066## ##STR00067## ##STR00068## ##STR00069## ##STR00070##

(71) In an embodiment, the organometallic compound may be electrically neutral.

(72) The organometallic compound represented by Formula 1 may include two metal atoms, and ligands coordinated with the metal atoms may be linked to each other, resulting in a rigid structure that leads to excellent molecular stability. Since the organometallic compound represented by Formula 1 has a structure in which angles of the ligands are tilted with respect to the metal atoms, excimer formation may be inhibited. Accordingly, a light-emitting device using the organometallic compound may have a low driving voltage and excellent luminance, efficiency, and lifespan.

(73) The organometallic compound may emit blue light. In an embodiment, the organometallic compound may emit blue light (bottom emission CIE.sub.x,y color coordinates 0.05 to 0.06) having a maximum emission wavelength in a range of about 430 nm to about 470 nm, but embodiments of the disclosure are not limited thereto. Accordingly, the organometallic compound represented by Formula 1 may be useful for the manufacture of a light-emitting device that emits blue light.

(74) Synthesis methods of the organometallic compound represented by Formula 1 may be recognizable by one of ordinary skill in the art by referring to Examples provided below.

(75) At least one of such organometallic compounds represented by Formula 1 may be used between a pair of electrodes of a light-emitting device. In an embodiment, the organometallic compound may be included in an emission layer. The organometallic compound included in the emission layer may act as a dopant. In embodiments, the organometallic compound represented by Formula 1 may be used as a material for a capping layer located outside a pair of electrodes of a light-emitting device.

(76) Accordingly, provided is a light-emitting device which may include a first electrode, a second electrode facing the first electrode, an interlayer disposed between the first electrode and the second electrode and including an emission layer, and at least one organometallic compound represented by Formula 1.

(77) The expression “(an interlayer) includes at least one of organometallic compounds” used herein may include a case in which “(an interlayer) includes identical organometallic compounds represented by Formula 1” and a case in which “(an interlayer) includes two or more different organometallic compounds represented by Formula 1.”

(78) In an embodiment, the interlayer may include, as the organometallic compound, only Compound BD1. In this regard, Compound BD1 may exist in the emission layer of the light-emitting device. In embodiments, the interlayer may include, as the organometallic compound, Compound BD1 and Compound BD2. In this regard, Compound BD1 and Compound BD2 may exist in an identical layer (for example, both Compound BD1 and Compound BD2 may exist in an emission layer), or different layers (for example, Compound BD1 may exist in an emission layer

and Compound BD2 may exist in an electron transport region).

(79) In an embodiment, the first electrode of the light-emitting device may be an anode, the second electrode of the light-emitting device may be a cathode, the interlayer may further include a hole transport region located between the first electrode and the emission layer and an electron transport region located between the emission layer and the second electrode, the hole transport region may include a hole injection layer, a hole transport layer, an emission auxiliary layer, an electron blocking layer, or any combination thereof, and the electron transport region may include a hole blocking layer, an electron transport layer, an electron injection layer, or any combination thereof.

(80) The term “interlayer” as used herein refers to a single layer and/or all layers between a first electrode and a second electrode of a light-emitting device. A material included in the “interlayer” is not limited to an organic material.

(81) In an embodiment, the emission layer may include the organometallic compound represented by Formula 1. In an embodiment, the emission layer may include a host and a dopant, and the dopant may include the organometallic compound represented by Formula 1. An amount of the host in the emission layer may be greater than an amount of the organometallic compound in the emission layer.

(82) In an embodiment, an amount of the host may be in a range of about 0.1 parts by weight to about 99 parts by weight, based on 100 parts by weight of the emission layer. In an embodiment, an amount of the host may be in a range of about 0.5 parts by weight to about 95 parts by weight, based on 100 parts by weight of the emission layer. For example, an amount of the host may be in a range of about 0.1 parts by weight to about 90 parts by weight, based on 100 parts by weight of the emission layer.

(83) The emission layer may emit red light, green light, blue light, and/or white light. In an embodiment, the emission layer may emit blue light having a maximum emission wavelength in a range of about 410 nm to about 500 nm.

(84) In an embodiment, the hole transport region may include an electron blocking layer, and the electron blocking layer may include the organometallic compound; or the electron transport region may include a hole blocking layer, and the hole blocking layer may include the organometallic compound.

(85) In an embodiment, the hole transport region may include a p-dopant having a lowest unoccupied molecular orbital (LUMO) energy level equal to or less than about -3.5 eV.

(86) In an embodiment, the electron transport region may include an electron transport layer and an electron injection layer, and at least one layer of the electron transport layer and the electron injection layer may include an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal compound, an alkaline earth-metal compound, a rare earth metal compound, an alkali metal complex, an alkaline earth-metal complex, a rare earth metal complex, or any combination thereof.

(87) Another aspect of the disclosure provides an electronic apparatus which may include the light-emitting device. The electronic apparatus may further include a thin-film transistor. In an embodiment, the electronic apparatus may further include a thin-film transistor including a source electrode and a drain electrode, and the first electrode of the light-emitting device may be electrically connected to the source electrode or the drain electrode. In an embodiment, the electronic apparatus may further include a color filter, a color conversion layer, a touch screen layer, a polarizing layer, or any combination thereof. More details on the electronic apparatus are the same as described in the specification.

(88) [Description of FIG. 1]

(89) FIG. 1 is a schematic cross-sectional view of a light-emitting device **10** according to an embodiment. The light-emitting device **10** includes a first electrode **110**, an interlayer **130**, and a second electrode **150**.

(90) Hereinafter, the structure of the light-emitting device **10** according to an embodiment and a method of manufacturing the light-emitting device **10** will be described in connection with FIG. 1.

(91) [First Electrode **110**]

(92) In FIG. **1**, a substrate may be further included under the first electrode **110** or above the second electrode **150**. The substrate may be a glass substrate or a plastic substrate. In embodiments, the substrate may be a flexible substrate, and may include plastics with excellent heat resistance and durability, such as polyimide, polyethylene terephthalate (PET), polycarbonate, polyethylene naphthalate, polyarylate (PAR), polyetherimide, or any combination thereof.

(93) The first electrode **110** may be formed by, for example, depositing or sputtering a material for forming the first electrode **110** on the substrate. When the first electrode **110** is an anode, a material for forming the first electrode **110** may be a high work function material that facilitates injection of holes.

(94) The first electrode **110** may be a reflective electrode, a semi-transmissive electrode, or a transmissive electrode. When the first electrode **110** is a transmissive electrode, a material for forming the first electrode **110** may include indium tin oxide (ITO), indium zinc oxide (IZO), tin oxide (SnO₂), zinc oxide (ZnO), or any combinations thereof. In embodiments, when the first electrode **110** is a semi-transmissive electrode or a reflective electrode, magnesium (Mg), silver (Ag), aluminum (Al), aluminum-lithium (Al—Li), calcium (Ca), magnesium-indium (Mg—In), magnesium-silver (Mg—Ag), or any combinations thereof may be used as a material for forming a first electrode **110**.

(95) The first electrode **110** may have a structure consisting of a single layer or a structure including multiple layers. In an embodiment, the first electrode **110** may have a three-layered structure of ITO/Ag/ITO.

(96) [Interlayer **130**]

(97) The interlayer **130** may be disposed on the first electrode **110**. The interlayer **130** may include an emission layer.

(98) The interlayer **130** may further include a hole transport region between the first electrode **110** and the emission layer and an electron transport region between the emission layer and the second electrode **150**.

(99) The interlayer **130** may further include, in addition to various organic materials, metal-containing compounds such as organometallic compounds, inorganic materials such as quantum dots, and the like.

(100) In embodiments, the interlayer **130** may include two or more light-emitting units sequentially stacked between the first electrode **110** and the second electrode **150** and a charge generation layer between the two or more light-emitting units. When the interlayer **130** includes the two or more light-emitting units and the charge generation layer as described above, the light-emitting device **10** may be a tandem light-emitting device.

(101) [Hole Transport Region in Interlayer **130**]

(102) The hole transport region may have a structure consisting of a layer consisting of a single material, a structure consisting of a layer consisting of different materials, or a multi-layered structure including layers including different materials.

(103) The hole transport region may include a hole injection layer, a hole transport layer, an emission auxiliary layer, an electron blocking layer, or any combination thereof.

(104) In an embodiment, the hole transport region may have a multi-layered structure including a hole injection layer/hole transport layer structure, a hole injection layer/hole transport layer/emission auxiliary layer structure, a hole injection layer/emission auxiliary layer structure, a hole transport layer/emission auxiliary layer structure, or a hole injection layer/hole transport layer/electron blocking layer structure, wherein, in each structure, layers may be stacked from the first electrode **110** in its respective stated order, but embodiments are not limited thereto.

(105) The hole transport region may include a compound represented by Formula 201, a compound represented by Formula 202, or any combination thereof:

(106) ##STR00071##

(107) In Formulae 201 and 202, L.sub.201 to L.sub.204 may each independently be a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, L.sub.205 may be *—O—*, *—S—*, *—N(Q.sub.201)—*, a C.sub.1-C.sub.20 alkylene group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.20 alkenylene group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, xa1 to xa4 may each independently be an integer from 0 to 5, xa5 may be an integer from 1 to 10, R.sub.201 to R.sub.204 and Q.sub.201 may each independently be a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, R.sub.201 and R.sub.202 may optionally be linked to each other, via a single bond, a C.sub.1-C.sub.5 alkylene group unsubstituted or substituted with at least one R.sub.10a, or a C.sub.2-C.sub.5 alkenylene group unsubstituted or substituted with at least one R.sub.10a, to form a C.sub.8-C.sub.60 polycyclic group (for example, a carbazole group or the like) unsubstituted or substituted with at least one R.sub.10a (for example, Compound HT16), R.sub.203 and R.sub.204 may optionally be linked to each other, via a single bond, a C.sub.1-C.sub.5 alkylene group unsubstituted or substituted with at least one R.sub.10a, or a C.sub.2-C.sub.5 alkenylene group unsubstituted or substituted with at least one R.sub.10a, to form a C.sub.8-C.sub.60 polycyclic group unsubstituted or substituted with at least one R.sub.10a, na1 may be an integer from 1 to 4, and * and *' each indicate a binding site to a neighboring atom. R.sub.10a may be the same as described in the specification.

(108) In an embodiment, each of Formulae 201 and 202 may include at least one of groups represented by Formulae CY201 to CY217:

(109) ##STR00072## ##STR00073##

(110) In Formulae CY201 to CY217, R.sub.10b and R.sub.10c may each independently be the same as described in connection with R.sub.10a, ring CY.sub.201 to ring CY.sub.204 may each independently be a C.sub.3-C.sub.20 carbocyclic group or a C.sub.1-C.sub.20 heterocyclic group, and at least one hydrogen in Formulae CY201 to CY217 may be unsubstituted or substituted with R.sub.10a. R.sub.10a may be the same as described in the specification.

(111) In an embodiment, ring CY.sub.201 to ring CY.sub.204 in Formulae CY201 to CY217 may each independently be a benzene group, a naphthalene group, a phenanthrene group, or an anthracene group.

(112) In embodiments, each of Formulae 201 and 202 may include at least one of groups represented by Formulae CY201 to CY203.

(113) In embodiments, Formula 201 may include at least one of groups represented by Formulae CY201 to CY203 and at least one of groups represented by Formulae CY204 to CY217.

(114) In embodiments, in Formula 201, xa1 may be 1, R.sub.201 may be a group represented by one of Formulae CY201 to CY203, xa2 may be 0, and R.sub.202 may be a group represented by one of Formulae CY204 to CY207.

(115) In embodiments, each of Formulae 201 and 202 may not include a group represented by one of Formulae CY201 to CY203.

(116) In embodiments, each of Formulae 201 and 202 may not include a group represented by one of Formulae CY201 to CY203, and may include at least one of groups represented by Formulae CY204 to CY217.

(117) In embodiments, each of Formulae 201 and 202 may not include a group represented by one of Formulae CY201 to CY217.

(118) In an embodiment, the hole transport region may include one of Compounds HT1 to HT46, m-MTDATA, TDATA, 2-TNATA, NPB(NPD), β -NPB, TPD, Spiro-TPD, Spiro-NPB, methylated-NPB, TAPC, HMTPD, 4,4',4''-tris(N-carbazolyl)triphenylamine (TCTA),

polyaniline/dodecylbenzenesulfonic acid (PANI/DBSA), poly(3,4-ethylenedioxythiophene)/poly(4-styrenesulfonate) (PEDOT/PSS), polyaniline/camphor sulfonic acid (PANI/CSA), polyaniline/poly(4-styrenesulfonate) (PANI/PSS), or any combination thereof:

(119) ##STR00074## ##STR00075## ##STR00076## ##STR00077## ##STR00078##

##STR00079## ##STR00080## ##STR00081## ##STR00082##

(120) A thickness of the hole transport region may be in a range of about 50 Å to about 10,000 Å. For example, the thickness of the hole transport region may be in a range of about 100 Å to about 4,000 Å. When the hole transport region includes a hole injection layer, a hole transport layer, or any combination thereof, a thickness of the hole injection layer may be in a range of about 100 Å to about 9,000 Å, and a thickness of the hole transport layer may be in a range of about 50 Å to about 2,000 Å. For example, the thickness of the hole injection layer may be in a range of 100 Å to about 1,000 Å. For example, the thickness of the hole transport layer may be in a range of about 100 Å to about 1,500 Å. When the thicknesses of the hole transport region, the hole injection layer, and the hole transport layer are within these ranges, satisfactory hole transporting characteristics may be obtained without a substantial increase in driving voltage.

(121) The emission auxiliary layer may increase light-emission efficiency by compensating for an optical resonance distance according to a wavelength of light emitted by an emission layer, and the electron blocking layer may block the leakage of electrons from an emission layer to a hole transport region. Materials that may be included in the hole transport region may be included in the emission auxiliary layer and the electron blocking layer.

(122) [p-Dopant]

(123) The hole transport region may further include, in addition to these materials, a charge-generation material for the improvement of conductive properties. The charge-generation material may be uniformly or non-uniformly dispersed in the hole transport region (for example, in the form of a single layer consisting of a charge-generation material).

(124) The charge-generation material may be, for example, a p-dopant.

(125) In an embodiment, a lowest unoccupied molecular orbital (LUMO) energy level of the p-dopant may be equal to or less than about -3.5 eV.

(126) In an embodiment, the p-dopant may include a quinone derivative, a cyano group-containing compound, a compound containing element EL1 and element EL2, or any combination thereof.

(127) Examples of the quinone derivative may include TCNQ and F4-TCNQ.

(128) Examples of the cyano group-containing compound may include HAT-CN and a compound represented by Formula 221.

(129) ##STR00083##

(130) In Formula 221, R.sub.221 to R.sub.223 may each independently be a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, wherein R.sub.10a may be the same as described in the specification, and at least one of R.sub.221 to R.sub.223 may each independently be a C.sub.3-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, each substituted with: a cyano group; —F; —Cl; —Br; —I; a C.sub.1-C.sub.20 alkyl group substituted with a cyano group, —F, —Cl, —Br, —I, or any combination thereof; or any combination thereof.

(131) In the compound containing element EL1 and element EL2, element EL1 may be a metal, a metalloid, or any combination thereof, and element EL2 may be a non-metal, a metalloid, or any combination thereof.

(132) Examples of the metal may include an alkali metal (for example, lithium (Li), sodium (Na), potassium (K), rubidium (Rb), cesium (Cs), etc.); an alkaline earth metal (for example, beryllium (Be), magnesium (Mg), calcium (Ca), strontium (Sr), barium (Ba), etc.); a transition metal (for example, titanium (Ti), zirconium (Zr), hafnium (Hf), vanadium (V), niobium (Nb), tantalum (Ta), chromium (Cr), molybdenum (Mo), tungsten (W), manganese (Mn), technetium (Tc), rhenium

(Re), iron (Fe), ruthenium (Ru), osmium (Os), cobalt (Co), rhodium (Rh), iridium (Ir), nickel (Ni), palladium (Pd), platinum (Pt), copper (Cu), silver (Ag), gold (Au), etc.); a post-transition metal (for example, zinc (Zn), indium (In), tin (Sn), etc.); and a lanthanide metal (for example, lanthanum (La), cerium (Ce), praseodymium (Pr), neodymium (Nd), promethium (Pm), samarium (Sm), europium (Eu), gadolinium (Gd), terbium (Tb), dysprosium (Dy), holmium (Ho), erbium (Er), thulium (Tm), ytterbium (Yb), lutetium (Lu), etc.).

(133) Examples of the metalloid may include silicon (Si), antimony (Sb), and tellurium (Te).

(134) Examples of the non-metal may include oxygen (O) and a halogen (for example, F, Cl, Br, I, etc.).

(135) In an embodiment, examples of the compound containing element EL1 and element EL2 may include a metal oxide, a metal halide (for example, metal fluoride, metal chloride, metal bromide, or metal iodide), a metalloid halide (for example, metalloid fluoride, metalloid chloride, metalloid bromide, or metalloid iodide), a metal telluride, or any combination thereof.

(136) Examples of the metal oxide may include tungsten oxide (for example, WO, W.sub.2O.sub.3, WO.sub.2, WO.sub.3, W.sub.2O.sub.5, etc.), vanadium oxide (for example, VO, V.sub.2O.sub.3, VO.sub.2, V.sub.2O.sub.5, etc.), molybdenum oxide (for example, MoO, Mo.sub.2O.sub.3, MoO.sub.2, MoO.sub.3, Mo.sub.2O.sub.5, etc.), and rhenium oxide (for example, ReO.sub.3, etc.).

(137) Examples of the metal halide may include an alkali metal halide, an alkaline earth metal halide, a transition metal halide, a post-transition metal halide, and a lanthanide metal halide.

(138) Examples of the alkali metal halide may include LiF, NaF, KF, RbF, CsF, LiCl, NaCl, KCl, RbCl, CsCl, LiBr, NaBr, KBr, RbBr, CsBr, LiI, NaI, KI, RbI, and CsI.

(139) Examples of the alkaline earth metal halide may include BeF.sub.2, MgF.sub.2, CaF.sub.2, SrF.sub.2, BaF.sub.2, BeCl.sub.2, MgCl.sub.2, CaCl.sub.2, SrCl.sub.2, BaCl.sub.2, BeBr.sub.2, MgBr.sub.2, CaBr.sub.2, SrBr.sub.2, BaBr.sub.2, BeI.sub.2, MgI.sub.2, CaI.sub.2, SrI.sub.2, and BaI.sub.2.

(140) Examples of the transition metal halide may include titanium halide (for example, TiF.sub.4, TiCl.sub.4, TiBr.sub.4, TiI.sub.4, etc.), zirconium halide (for example, ZrF.sub.4, ZrCl.sub.4, ZrBr.sub.4, ZrI.sub.4, etc.), hafnium halide (for example, HfF.sub.4, HfCl.sub.4, HfBr.sub.4, HfI.sub.4, etc.), vanadium halide (for example, VF.sub.3, VCl.sub.3, VBr.sub.3, VI.sub.3, etc.), niobium halide (for example, NbF.sub.3, NbCl.sub.3, NbBr.sub.3, NbI.sub.3, etc.), tantalum halide (for example, TaF.sub.3, TaCl.sub.3, TaBr.sub.3, TaI.sub.3, etc.), chromium halide (for example, CrF.sub.3, CrCl.sub.3, CrBr.sub.3, CrI.sub.3, etc.), molybdenum halide (for example, MoF.sub.3, MoCl.sub.3, MoBr.sub.3, MoI.sub.3, etc.), tungsten halide (for example, WF.sub.3, WCl.sub.3, WBr.sub.3, WI.sub.3, etc.), manganese halide (for example, MnF.sub.2, MnCl.sub.2, MnBr.sub.2, MnI.sub.2, etc.), technetium halide (for example, TcF.sub.2, TcCl.sub.2, TcBr.sub.2, TcI.sub.2, etc.), rhenium halide (for example, ReF.sub.2, ReCl.sub.2, ReBr.sub.2, ReI.sub.2, etc.), iron halide (for example, FeF.sub.2, FeCl.sub.2, FeBr.sub.2, FeI.sub.2, etc.), ruthenium halide (for example, RuF.sub.2, RuCl.sub.2, RuBr.sub.2, RuI.sub.2, etc.), osmium halide (for example, OsF.sub.2, OsCl.sub.2, OsBr.sub.2, OsI.sub.2, etc.), cobalt halide (for example, CoF.sub.2, CoCl.sub.2, CoBr.sub.2, CoI.sub.2, etc.), rhodium halide (for example, RhF.sub.2, RhCl.sub.2, RhBr.sub.2, RhI.sub.2, etc.), iridium halide (for example, IrF.sub.2, IrCl.sub.2, IrBr.sub.2, IrI.sub.2, etc.), nickel halide (for example, NiF.sub.2, NiCl.sub.2, NiBr.sub.2, NiI.sub.2, etc.), palladium halide (for example, PdF.sub.2, PdCl.sub.2, PdBr.sub.2, PdI.sub.2, etc.), platinum halide (for example, PtF.sub.2, PtCl.sub.2, PtBr.sub.2, PtI.sub.2, etc.), copper halide (for example, CuF, CuCl, CuBr, CuI, etc.), silver halide (for example, AgF, AgCl, AgBr, AgI, etc.), and gold halide (for example, AuF, AuCl, AuBr, AuI, etc.).

(141) Examples of the post-transition metal halide may include zinc halide (for example, ZnF.sub.2, ZnCl.sub.2, ZnBr.sub.2, ZnI.sub.2, etc.), indium halide (for example, InI.sub.3, etc.), and tin halide (for example, SnI.sub.2, etc.).

(142) Examples of the lanthanide metal halide may include YbF, YbF.sub.2, YbF.sub.3, SmF.sub.3,

YbCl, YbCl.sub.2, YbCl.sub.3 SmCl.sub.3, YbBr, YbBr.sub.2, YbBr.sub.3 SmBr.sub.3, YbI, YbI.sub.2, YbI.sub.3, and SmI.sub.3.

(143) Examples of the metalloid halide may include antimony halide (for example, SbCl.sub.5, etc.).

(144) Examples of the metal telluride may include an alkali metal telluride (for example, Li.sub.2Te, Na.sub.2Te, K.sub.2Te, Rb.sub.2Te, Cs.sub.2Te, etc.), an alkaline earth metal telluride (for example, BeTe, MgTe, CaTe, SrTe, BaTe, etc.), a transition metal telluride (for example, TiTe.sub.2, ZrTe.sub.2, HfTe.sub.2, V.sub.2Te.sub.3, Nb.sub.2Te.sub.3, Ta.sub.2Te.sub.3, Cr.sub.2Te.sub.3, Mo.sub.2Te.sub.3, W.sub.2Te.sub.3, MnTe, TcTe, ReTe, FeTe, RuTe, OsTe, CoTe, RhTe, IrTe, NiTe, PdTe, PtTe, Cu.sub.2Te, CuTe, Ag.sub.2Te, AgTe, Au.sub.2Te, etc.), a post-transition metal telluride (for example, ZnTe, etc.), and a lanthanide metal telluride (for example, LaTe, CeTe, PrTe, NdTe, PmTe, EuTe, GdTe, TbTe, DyTe, HoTe, ErTe, TmTe, YbTe, LuTe, etc.).

(145) [Emission Layer in Interlayer **130**]

(146) The light-emitting device according to an embodiment may include an emission layer in an interlayer. The emission layer may be the same as described above, or as described in connection with the light-emitting device below.

(147) When the light-emitting device **10** is a full-color light-emitting device, the emission layer may be patterned into a red emission layer, a green emission layer, and/or a blue emission layer, according to a subpixel. In embodiments, the emission layer may have a stacked structure of two or more layers of a red emission layer, a green emission layer, and a blue emission layer, in which the two or more layers may contact each other or may be separated from each other. In embodiments, the emission layer may include two or more materials of a red light-emitting material, a green light-emitting material, and a blue light-emitting material, in which the two or more materials are mixed with each other in a single layer to emit white light.

(148) The emission layer may include a host and a dopant. The dopant may include a phosphorescent dopant, a fluorescent dopant, or any combination thereof.

(149) An amount of the dopant in the emission layer may be in a range of about 0.01 to about 15 parts by weight, based on 100 parts by weight of the host.

(150) In embodiments, the emission layer may include a quantum dot.

(151) The emission layer may include a delayed fluorescence material. The delayed fluorescence material may act as a host or a dopant in the emission layer.

(152) A thickness of the emission layer may be in a range of about 100 Å to about 1,000 Å. For example, the thickness of the emission layer may be in a range of about 200 Å to about 600 Å. When the thickness of the emission layer is within this range, excellent light-emission characteristics may be obtained without a substantial increase in driving voltage.

(153) [Host]

(154) The host may include a compound represented by Formula 301:

[Ar.sub.301].sub.xb11-[(L.sub.301).sub.xb1-R.sub.301].sub.xb21 [Formula 301]

(155) In Formula 301, Ar.sub.301 and L.sub.301 may each independently be a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, xb11 may be 1, 2, or 3, xb1 may be an integer from 0 to 5, R.sub.301 may be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.60 alkyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkenyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkynyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 alkoxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, —Si(Q.sub.301)(Q.sub.302)(Q.sub.303), —N(Q.sub.301)

(Q.sub.302), —B(Q.sub.301)(Q.sub.302), —C(=O)(Q.sub.301), —S(=O).sub.2(Q.sub.301), or —P(=O)(Q.sub.301)(Q.sub.302), xb21 may be an integer from 1 to 5, and Q.sub.301 to Q.sub.303 may each independently be the same as described in connection with Q.sub.1. R.sub.10a may be the same as described in the specification.

(156) In an embodiment, when xb11 in Formula 301 is 2 or more, two or more of Ar.sub.301(s) may be linked to each other via a single bond.

(157) In embodiments, the host may include a compound represented by Formula 301-1, a compound represented by Formula 301-2, or any combination thereof:

(158) ##STR00084##

(159) In Formulae 301-1 and 301-2,

(160) ring A.sub.301 to ring A.sub.304 may each independently be a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, X.sub.301 may be O, S, N—[(L.sub.304).sub.xb4-R.sub.304], C(R.sub.304)(R.sub.305), or Si(R.sub.304)(R.sub.305), xb22 and xb23 may each independently be 0, 1, or 2, L.sub.301, xb1, and R.sub.301 may each respectively be the same as described in the specification, L.sub.302 to L.sub.304 may each independently be the same as described in connection with L.sub.301, xb2 to xb4 may each independently be the same as described in connection with xb1, and R.sub.302 to R.sub.305 and R.sub.311 to R.sub.314 may each independently be the same as described in connection with R.sub.301. R.sub.10a may be the same as described in the specification.

(161) In embodiments, the host may include an alkali earth metal complex, a post-transition metal complex, or any combination thereof. In an embodiment, the host may include a Be complex (for example, Compound H55), a Mg complex, a Zn complex, or any combination thereof.

(162) In embodiments, the host may include one of Compounds H1 to H124, 9,10-di(2-naphthyl)anthracene (ADN), 2-methyl-9,10-bis(naphthalen-2-yl)anthracene (MADN), 9,10-di(2-naphthyl)-2-t-butyl-anthracene (TBADN), 4,4'-bis(N-carbazolyl)-1,1'-biphenyl (CBP), 1,3-di-9-carbazolylbenzene (mCP), 1,3,5-tri(carbazol-9-yl)benzene (TCP), or any combination thereof:

(163) ##STR00085## ##STR00086## ##STR00087## ##STR00088## ##STR00089##
##STR00090## ##STR00091## ##STR00092## ##STR00093## ##STR00094## ##STR00095##
##STR00096## ##STR00097## ##STR00098## ##STR00099## ##STR00100## ##STR00101##
##STR00102## ##STR00103## ##STR00104## ##STR00105## ##STR00106## ##STR00107##
##STR00108## ##STR00109## ##STR00110## ##STR00111## ##STR00112## ##STR00113##
##STR00114## ##STR00115## ##STR00116## ##STR00117##

[Phosphorescent Dopant]

(164) The phosphorescent dopant may include at least one transition metal as a central metal.

(165) The phosphorescent dopant may include a monodentate ligand, a bidentate ligand, a tridentate ligand, a tetradentate ligand, a pentadentate ligand, a hexadentate ligand, or any combination thereof.

(166) The phosphorescent dopant may be electrically neutral.

(167) In an embodiment, the phosphorescent dopant may include an organometallic compound represented by Formula 401:

(168) ##STR00118##

(169) In Formulae 401 and 402, M may be a transition metal (for example, iridium (Ir), platinum (Pt), palladium (Pd), osmium (Os), titanium (Ti), gold (Au), hafnium (Hf), europium (Eu), terbium (Tb), rhodium (Rh), rhenium (Re), or thulium (Tm)), L.sub.401 may be a ligand represented by Formula 402, and xc1 may be 1, 2, or 3, wherein when xc1 is 2 or more, two or more of L.sub.401(s) may be identical to or different from each other, L.sub.402 may be an organic ligand, and xc2 may be 0, 1, 2, 3, or 4, wherein when xc2 is 2 or more, two or more of L.sub.402(s) may be identical to or different from each other, X.sub.401 and X.sub.402 may each independently be nitrogen (N) or carbon (C), ring A.sub.401 and ring A.sub.402 may each independently be a

C.sub.3-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, T.sub.401 may be a single bond, *—O—*, *—S—*, *—C(=O)—*, *—N(Q.sub.411)—*, *—C(Q.sub.411)(Q.sub.412)—*, *—C(Q.sub.411)=C(Q.sub.412)—*, *—C(Q.sub.411)=*, or *=C=*, X.sub.403 and X.sub.404 may each independently be a chemical bond (for example, a covalent bond or a coordination bond), O, S, N(Q.sub.413), B(Q.sub.413), P(Q.sub.413), C(Q.sub.413)(Q.sub.414), or Si(Q.sub.413)(Q.sub.414), Q.sub.411 to Q.sub.414 may each independently be the same as described in connection with Q.sub.1, R.sub.401 and R.sub.402 may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.20 alkyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.20 alkoxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, —Si(Q.sub.401)(Q.sub.402)(Q.sub.403), —N(Q.sub.401)(Q.sub.402), —B(Q.sub.401)(Q.sub.402), —C(=O)(Q.sub.401), —S(=O).sub.2(Q.sub.401), or —P(=O)(Q.sub.401)(Q.sub.402), Q.sub.401 to Q.sub.403 may each independently be the same as described in connection with Q.sub.1, xc11 and xc12 may each independently be an integer from 0 to 10, and * and *' in Formula 402 each indicates a binding site to M in Formula 401. R.sub.10a may be the same as described in the specification.

(170) In an embodiment, in Formula 402, X.sub.401 may be nitrogen and X.sub.402 may be carbon, or each of X.sub.401 and X.sub.402 may be nitrogen.

(171) In embodiments, when xc1 in Formula 402 is 2 or more, two ring A.sub.401(s) in two or more of L.sub.401(s) may optionally be linked to each other via T.sub.402, which is a linking group, and two ring A.sub.402(s) may optionally be linked to each other via T.sub.403, which is a linking group (see Compounds PD1 to PD4 and PD7). T.sub.402 and T.sub.403 may each independently be the same as described in connection with T.sub.401.

(172) L.sub.402 in Formula 401 may be an organic ligand. In an embodiment, L.sub.402 may include a halogen group, a diketone group (for example, an acetylacetonate group), a carboxylic acid group (for example, a picolinate group), —C(=O), an isonitrile group, —CN, a phosphorus-containing group (for example, a phosphine group, a phosphite group, etc.), or any combination thereof.

(173) The phosphorescent dopant may include, for example, one of Compounds PD1 to PD25, or any combination thereof:

(174) ##STR00119## ##STR00120## ##STR00121## ##STR00122## ##STR00123##
[Fluorescent Dopant]

(175) The fluorescent dopant may include an amine group-containing compound, a styryl group-containing compound, or any combination thereof.

(176) In an embodiment, the fluorescent dopant may include a compound represented by Formula 501:

(177) ##STR00124##

(178) In Formula 501, Ar.sub.501, L.sub.501 to L.sub.503, R.sub.501, and R.sub.502 may each independently be a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, xd1 to xd3 may each independently be 0, 1, 2, or 3, and xd4 may be 1, 2, 3, 4, 5, or 6.

(179) In an embodiment, Ar.sub.501 in Formula 501 may be a condensed cyclic group (for example, an anthracene group, a chrysene group, or a pyrene group) in which three or more monocyclic groups are condensed together.

(180) In embodiments, xd4 in Formula 501 may be 2.

(181) In an embodiment, the fluorescent dopant may include one of Compounds FD1 to FD36, DPVBi, DPAVBi, or any combination thereof:

(182) ##STR00125## ##STR00126## ##STR00127## ##STR00128## ##STR00129##

[Delayed Fluorescence Material]

(183) The emission layer may include a delayed fluorescence material.

(184) In the specification, the delayed fluorescence material may be selected from compounds capable of emitting delayed fluorescence based on a delayed fluorescence emission mechanism.

(185) The delayed fluorescence material included in the emission layer may act as a host or a dopant depending on the type of other materials included in the emission layer.

(186) In an embodiment, a difference between a triplet energy level (eV) of the delayed fluorescence material and a singlet energy level (eV) of the delayed fluorescence material may be in a range of about 0 eV to about 0.5 eV. When the difference between the triplet energy level (eV) of the delayed fluorescence material and the singlet energy level (eV) of the delayed fluorescence material satisfies the above-described range, up-conversion from the triplet state to the singlet state of the delayed fluorescence material may effectively occur, and thus, the luminescence efficiency of the light-emitting device **10** may be improved.

(187) In an embodiment, the delayed fluorescence material may include a material including at least one electron donor (for example, a π electron-rich C.sub.3-C.sub.60 cyclic group, such as a carbazole group) and at least one electron acceptor (for example, a sulfoxide group, a cyano group, or a π electron-deficient nitrogen-containing C.sub.1-C.sub.60 cyclic group), and a material including a C.sub.8-C.sub.60 polycyclic group in which two or more cyclic groups are condensed while sharing boron (B).

(188) Examples of the delayed fluorescence material may include at least one of Compounds DF1 to DF9:

(189) ##STR00130## ##STR00131##

[Quantum Dot]

(190) The emission layer may include a quantum dot.

(191) In the specification, a quantum dot may be a crystal of a semiconductor compound, and may include any material capable of emitting light of various emission wavelengths according to the size of the crystal.

(192) A diameter of the quantum dot may be, for example, in a range of about 1 nm to about 10 nm.

(193) The quantum dot may be synthesized by a wet chemical process, a metal organic chemical vapor deposition process, a molecular beam epitaxy process, or any process similar thereto.

(194) According to the wet chemical process, a precursor material is mixed with an organic solvent to grow a quantum dot particle crystal. When the crystal grows, the organic solvent naturally acts as a dispersant coordinated on the surface of the quantum dot crystal and controls the growth of the crystal so that the growth of quantum dot particles can be controlled through a process which is more easily performed than vapor deposition methods, such as metal organic chemical vapor deposition (MOCVD) or molecular beam epitaxy (MBE), and which requires low costs.

(195) The quantum dot may include Group II-VI semiconductor compounds, Group III-V semiconductor compounds, Group III-VI semiconductor compounds, Group I-III-VI semiconductor compounds, Group IV-VI semiconductor compounds, a Group IV element or compound, or any combination thereof.

(196) Examples of the Group II-VI semiconductor compound may include a binary compound, such as CdSe, CdTe, ZnS, ZnSe, ZnTe, ZnO, HgS, HgSe, HgTe, MgSe, or MgS; a ternary compound, such as CdSeS, CdSeTe, CdSTe, ZnSeS, ZnSeTe, ZnSTe, HgSeS, HgSeTe, HgSTe, CdZnS, CdZnSe, CdZnTe, CdHgS, CdHgSe, CdHgTe, HgZnS, HgZnSe, HgZnTe, MgZnSe, or MgZnS; a quaternary compound, such as CdZnSeS, CdZnSeTe, CdZnSTe, CdHgSeS, CdHgSeTe, CdHgSTe, HgZnSeS, HgZnSeTe, or HgZnSTe; or any combination thereof.

(197) Examples of the Group III-V semiconductor compound may include a binary compound, such as GaN, GaP, GaAs, GaSb, AlN, AlP, AlAs, AlSb, InN, InP, InAs, or InSb; a ternary compound, such as GaNP, GaNAs, GaNSb, GaPAs, GaPSb, AlNP, AlNAs, AlNSb, AlPAs, AlPSb,

InGaP, InNP, InAlP, InNAs, InNSb, InPAs, or InPSb; a quaternary compound, such as GaAlNP, GaAlNAs, GaAlNSb, GaAlPAs, GaAlPSb, GaInNP, GaInNAs, GaInNSb, GaInPAs, GaInPSb, InAlNP, InAlNAs, InAlNSb, InAlPAs, or InAlPSb; or any combination thereof. The Group III-V semiconductor compound may further include a Group II element. Examples of the Group III-V semiconductor compound further including a Group II element may include InZnP, InGaZnP, or InAlZnP.

(198) Examples of the Group III-VI semiconductor compound may include a binary compound, such as GaS, GaSe, Ga.sub.2Se.sub.3, GaTe, InS, InSe, In.sub.2S.sub.3, In.sub.2Se.sub.3, or InTe; a ternary compound, such as InGaS.sub.3 or InGaSe.sub.3; or any combination thereof.

(199) Examples of the Group I-III-VI semiconductor compound may include: a ternary compound, such as AgInS, AgInS.sub.2, CuInS, CuInS.sub.2, CuGaO.sub.2, AgGaO.sub.2, or AgAlO.sub.2; or any combination thereof.

(200) Examples of the Group IV-VI semiconductor compound may include a binary compound, such as SnS, SnSe, SnTe, PbS, PbSe, or PbTe; a ternary compound, such as SnSeS, SnSeTe, SnSTe, PbSeS, PbSeTe, PbSTe, SnPbS, SnPbSe, or SnPbTe; a quaternary compound, such as SnPbSSe, SnPbSeTe, or SnPbSTe; or any combination thereof.

(201) The Group IV element or compound may include a single element material, such as Si or Ge; a binary compound, such as SiC or SiGe; or any combination thereof.

(202) Each element included in a multi-element compound such as the binary compound, the ternary compound, and the quaternary compound, may exist at a uniform concentration or at a non-uniform concentration in a particle.

(203) The quantum dot may have a single structure or a core-shell structure. In the case of the quantum dot having a single structure, the concentration of each element included in the corresponding quantum dot may be uniform. In an embodiment, the material included in the core and the material included in the shell may be different from each other.

(204) The shell of the quantum dot may be a protective layer that prevents chemical degeneration of the core to maintain semiconductor characteristics and/or may be a charging layer that imparts electrophoretic characteristics to the quantum dot. The shell may be a single layer or a multi-layer. An interface between the core and the shell may have a concentration gradient in which the concentration of an element existing in the shell decreases toward the center of the core.

(205) Examples of the shell of the quantum dot may include an oxide of metal, an oxide of a metalloid, an oxide of a non-metal, a semiconductor compound, or a combination thereof. Examples of the oxide of metal, metalloid, or non-metal may include a binary compound, such as SiO.sub.2, Al.sub.2O.sub.3, TiO.sub.2, ZnO, MnO, Mn.sub.2O.sub.3, Mn.sub.3O.sub.4, CuO, FeO, Fe.sub.2O.sub.3, Fe.sub.3O.sub.4, CoO, Co.sub.3O.sub.4, or NiO; a ternary compound, such as MgAl.sub.2O.sub.4, CoFe.sub.2O.sub.4, NiFe.sub.2O.sub.4, or CoMn.sub.2O.sub.4; or any combination thereof. Examples of the semiconductor compound may include, as described herein, Group II-VI semiconductor compounds; Group III-V semiconductor compounds; Group III-VI semiconductor compounds; Group I-III-VI semiconductor compounds; Group IV-VI semiconductor compounds; or any combination thereof. In an embodiment, the semiconductor compound may include CdS, CdSe, CdTe, ZnS, ZnSe, ZnTe, ZnSeS, ZnTeS, GaAs, GaP, GaSb, HgS, HgSe, HgTe, InAs, InP, InGaP, InSb, AlAs, AlP, AlSb, or any combination thereof.

(206) A full width at half maximum (FWHM) of an emission wavelength spectrum of the quantum dot may be equal to or less than about 45 nm. For example, a FWHM of an emission wavelength spectrum of the quantum dot may be equal to or less than about 40 nm. For example, a FWHM of an emission wavelength spectrum of the quantum dot may be equal to or less than about 30 nm. Within these ranges, color purity or color gamut may be increased. Light emitted through the quantum dot may be emitted in all directions, and a wide viewing angle may be improved.

(207) The quantum dot may be a spherical particle, a pyramidal particle, a multi-arm particle, a cubic nanoparticle, a nanotube particle, a nanowire particle, a nanofiber particle, or a nanoplate

particle.

(208) Since the energy band gap may be adjusted by controlling the size of the quantum dot, light having various wavelength bands may be obtained from the quantum dot emission layer.

Accordingly, by using quantum dots of different sizes, a light-emitting device that emits light of various wavelengths may be implemented. For example, the size of the quantum dot may be selected to emit red, green and/or blue light. The size of the quantum dot may be configured to emit white light by combining light of various colors.

(209) [Electron Transport Region in Interlayer **130**]

(210) The electron transport region may have a structure consisting of a layer consisting of a single material, a structure consisting of a layer consisting of different materials, or a multi-layered structure including layers including different materials.

(211) The electron transport region may include a buffer layer, a hole blocking layer, an electron control layer, an electron transport layer, an electron injection layer, or any combination thereof.

(212) In an embodiment, the electron transport region may have an electron transport layer/electron injection layer structure, a hole blocking layer/electron transport layer/electron injection layer structure, an electron control layer/electron transport layer/electron injection layer structure, or a buffer layer/electron transport layer/electron injection layer structure, wherein, for each structure, constituting layers may be stacked from an emission layer in its respective stated order, but embodiments are not limited thereto.

(213) The electron transport region (for example, the buffer layer, the hole blocking layer, the electron control layer, or the electron transport layer in the electron transport region) may include a metal-free compound including at least one π electron-deficient nitrogen-containing C.sub.1-C.sub.60 cyclic group.

(214) In an embodiment, the electron transport region may include a compound represented by Formula 601:

[Ar.sub.601].sub.xe11-[(L.sub.601).sub.xe1-R.sub.601].sub.xe21 [Formula 601]

(215) In Formula 601, Ar.sub.601 and L.sub.601 may each independently be a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, xe11 may be 1, 2, or 3, xe1 may be 0, 1, 2, 3, 4, or 5, R.sub.601 may be a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, —Si(Q.sub.601)(Q.sub.602)(Q.sub.603), —C(=O)(Q.sub.601), —S(=O).sub.2(Q.sub.601), or —P(=O)(Q.sub.601)(Q.sub.602), Q.sub.601 to Q.sub.603 may each independently be the same as described in connection with Q.sub.1, xe21 may be 1, 2, 3, 4, or 5, and at least one of Ar.sub.601, L.sub.601, and R.sub.601 may each independently be a π electron-deficient nitrogen-containing C.sub.1-C.sub.60 cyclic group unsubstituted or substituted with at least one R.sub.10a. R.sub.10a may be the same as described in the specification.

(216) In an embodiment, when xe11 in Formula 601 is 2 or more, two or more of Ar.sub.601(s) may be linked to each other via a single bond.

(217) In embodiments, Ar.sub.601 in Formula 601 may be a substituted or unsubstituted anthracene group.

(218) In embodiments, the electron transport region may include a compound represented by Formula 601-1:

(219) ##STR00132##

(220) In Formula 601-1, X.sub.614 may be N or C(R.sub.614), X.sub.615 may be N or C(R.sub.615), X.sub.616 may be N or C(R.sub.616), and at least one of X.sub.614 to X.sub.616 may be N, L.sub.611 to L.sub.613 may each independently be the same as described in connection with L.sub.601, xe611 to xe613 may each independently be the same as described in connection with xe1, R.sub.611 to R.sub.613 may each independently be the same as described in connection

with R.sub.601, and R.sub.614 to R.sub.616 may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a. R.sub.10a may be the same as described in the specification.

(221) In an embodiment, xe1 and xe611 to xe613 in Formulae 601 and 601-1 may each independently be 0, 1, or 2.

(222) The electron transport region may include one of Compounds ET1 to ET45, 2,9-dimethyl-4,7-diphenyl-1,10-phenanthroline (BCP), 4,7-diphenyl-1,10-phenanthroline (Bphen), Alq.sub.3, BALq, TAZ, NTAZ, or any combination thereof:

(223) ##STR00133## ##STR00134## ##STR00135## ##STR00136## ##STR00137##
##STR00138## ##STR00139## ##STR00140## ##STR00141## ##STR00142## ##STR00143##
##STR00144## ##STR00145## ##STR00146## ##STR00147##

(224) A thickness of the electron transport region may be in a range of about 100 Å to about 5,000 Å. For example, the thickness of the electron transport region may be in a range of about 160 Å to about 4,000 Å. When the electron transport region includes a buffer layer, a hole blocking layer, an electron control layer, an electron transport layer, or any combination thereof, a thickness of the buffer layer, the hole blocking layer, or the electron control layer may each independently be in a range of about 20 Å to about 1,000 Å, and a thickness of the electron transport layer may be in a range of about 100 Å to about 1,000 Å. For example, the thickness of the buffer layer, the hole blocking layer, or the electron control layer may each independently be in a range of about 30 Å to about 300 Å. For example, the thickness of the electron transport layer may be in a range of about 150 Å to about 500 Å. When the thicknesses of the buffer layer, the hole blocking layer, the electron control layer, and/or the electron transport layer are within these ranges, satisfactory electron transporting characteristics may be obtained without a substantial increase in driving voltage.

(225) The electron transport region (for example, the electron transport layer in the electron transport region) may further include, in addition to the materials described above, a metal-containing material.

(226) The metal-containing material may include an alkali metal complex, an alkaline earth metal complex, or any combination thereof. A metal ion of the alkali metal complex may be a Li ion, a Na ion, a K ion, a Rb ion, or a Cs ion, and a metal ion of the alkaline earth metal complex may be a Be ion, a Mg ion, a Ca ion, a Sr ion, or a Ba ion. A ligand coordinated with the metal ion of the alkali metal complex or the alkaline earth-metal complex may include a hydroxyquinoline, a hydroxyisoquinoline, a hydroxybenzoquinoline, a hydroxyacridine, a hydroxyphenanthridine, a hydroxyphenyloxazole, a hydroxyphenylthiazole, a hydroxyphenyloxadiazole, a hydroxyphenylthiadiazole, a hydroxyphenylpyridine, a hydroxyphenylbenzimidazole, a hydroxyphenylbenzothiazole, a bipyridine, a phenanthroline, a cyclopentadiene, or any combination thereof.

(227) In an embodiment, the metal-containing material may include a Li complex. The Li complex may include, for example, Compound ET-D1 (LiQ) or ET-D2:

(228) ##STR00148##

(229) The electron transport region may include an electron injection layer that facilitates the injection of electrons from the second electrode **150**. The electron injection layer may directly contact the second electrode **150**.

(230) The electron injection layer may have a structure consisting of a layer consisting of a single material, a structure consisting of a layer consisting of different materials, or a multi-layered structure including layers including different materials.

(231) The electron injection layer may include an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal-containing compound, an alkaline earth metal-containing compound, a rare

earth metal-containing compound, an alkali metal complex, an alkaline earth metal complex, a rare earth metal complex, or any combination thereof.

(232) The alkali metal may include Li, Na, K, Rb, Cs, or any combination thereof. The alkaline earth metal may include Mg, Ca, Sr, Ba, or any combination thereof. The rare earth metal may include Sc, Y, Ce, Tb, Yb, Gd, or any combination thereof.

(233) The alkali metal-containing compound, the alkaline earth metal-containing compound, and the rare earth metal-containing compound may include oxides, halides (for example, fluorides, chlorides, bromides, or iodides), or tellurides of the alkali metal, the alkaline earth metal, and the rare earth metal, or any combination thereof.

(234) The alkali metal-containing compound may include alkali metal oxides, such as Li_2O , Cs_2O , or K_2O , alkali metal halides, such as LiF , NaF , CsF , KF , LiI , NaI , CsI , or KI , or any combination thereof. The alkaline earth metal-containing compound may include an alkaline earth metal oxides, such as BaO , SrO , CaO , $\text{Ba}_x\text{Sr}_{1-x}\text{O}$ (x is a real number satisfying the condition of $0 < x < 1$), $\text{Ba}_x\text{Ca}_{1-x}\text{O}$ (x is a real number satisfying the condition of $0 < x < 1$), or the like. The rare earth metal-containing compound may include YbF_3 , ScF_3 , Sc_2O_3 , Y_2O_3 , Ce_2O_3 , GdF_3 , TbF_3 , YbI_3 , ScI_3 , TbI_3 , or any combination thereof. In embodiments, the rare earth metal-containing compound may include a lanthanide metal telluride. Examples of the lanthanide metal telluride may include LaTe , CeTe , PrTe , NdTe , PmTe , SmTe , EuTe , GdTe , TbTe , DyTe , HoTe , ErTe , TmTe , YbTe , LuTe , La_2Te_3 , Ce_2Te_3 , Pr_2Te_3 , Nd_2Te_3 , Pm_2Te_3 , Sm_2Te_3 , Eu_2Te_3 , Gd_2Te_3 , Tb_2Te_3 , Dy_2Te_3 , Ho_2Te_3 , Er_2Te_3 , Tm_2Te_3 , Yb_2Te_3 , and Lu_2Te_3 .

(235) The alkali metal complex, the alkaline earth-metal complex, and the rare earth metal complex may include one of ions of the alkali metal, ions of the alkaline earth metal, and ions of the rare earth metal and a ligand bonded to the metal ion, for example, hydroxyquinoline, hydroxyisoquinoline, hydroxybenzoquinoline, hydroxyacridine, hydroxyphenanthridine, hydroxyphenyloxazole, hydroxyphenylthiazole, hydroxyphenyloxadiazole, hydroxyphenylthiadiazole, hydroxyphenylpyridine, hydroxyphenyl benzimidazole, hydroxyphenylbenzothiazole, bipyridine, phenanthroline, cyclopentadiene, or any combination thereof.

(236) The electron injection layer may consist of an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal-containing compound, an alkaline earth metal-containing compound, a rare earth metal-containing compound, an alkali metal complex, an alkaline earth metal complex, a rare earth metal complex, or any combination thereof, as described above. In embodiments, the electron injection layer may further include an organic material (for example, a compound represented by Formula 601).

(237) In an embodiment, the electron injection layer may consist of an alkali metal-containing compound (for example, an alkali metal halide); or an alkali metal-containing compound (for example, an alkali metal halide), and an alkali metal, an alkaline earth metal, a rare earth metal, or any combination thereof. In an embodiment, the electron injection layer may be a $\text{KI}:\text{Yb}$ co-deposited layer, an $\text{RbI}:\text{Yb}$ co-deposited layer, or the like.

(238) When the electron injection layer further includes an organic material, an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal-containing compound, an alkaline earth metal-containing compound, a rare earth metal-containing compound, an alkali metal complex, an alkaline earth-metal complex, a rare earth metal complex, or any combination thereof may be homogeneously or non-homogeneously dispersed in a matrix including the organic material.

(239) A thickness of the electron injection layer may be in a range of about 1 Å to about 100 Å. For example, the thickness of the electron injection layer may be in a range of about 3 Å to about 90 Å. When the thickness of the electron injection layer is within this range, satisfactory electron injection characteristics may be obtained without a substantial increase in driving voltage.

(240) [Second Electrode **150**]

(241) The second electrode **150** may be disposed on the interlayer **130** having such a structure. The second electrode **150** may be a cathode, which is an electron injection electrode, and as the material for the second electrode **150**, a metal, an alloy, an electrically conductive compound, or any combination thereof, each having a low work function, may be used.

(242) The second electrode **150** may include lithium (Li), silver (Ag), magnesium (Mg), aluminum (Al), aluminum-lithium (Al—Li), calcium (Ca), magnesium-indium (Mg—In), magnesium-silver (Mg—Ag), ytterbium (Yb), silver-ytterbium (Ag—Yb), ITO, IZO, or any combination thereof. The second electrode **150** may be a transmissive electrode, a semi-transmissive electrode, or a reflective electrode.

(243) The second electrode **150** may have a single-layered structure or a multi-layered structure including two or more layers.

(244) [Capping Layer]

(245) A first capping layer may be located outside the first electrode **110**, and/or a second capping layer may be located outside the second electrode **150**. For example, the light-emitting device **10** may have a structure in which the first capping layer, the first electrode **110**, the interlayer **130**, and the second electrode **150** are stacked in this stated order, a structure in which the first electrode **110**, the interlayer **130**, the second electrode **150**, and the second capping layer are stacked in this stated order, or a structure in which the first capping layer, the first electrode **110**, the interlayer **130**, the second electrode **150**, and the second capping layer are stacked in this stated order.

(246) Light generated in an emission layer of the interlayer **130** of the light-emitting device **10** may be extracted toward the outside through the first electrode **110** (which may be a semi-transmissive electrode or a transmissive electrode) and through the first capping layer. Light generated in an emission layer of the interlayer **130** of the light-emitting device **10** may be extracted toward the outside through the second electrode **150** (which may be a semi-transmissive electrode or a transmissive electrode) and through the second capping layer.

(247) The first capping layer and the second capping layer may each increase external luminescence efficiency according to the principle of constructive interference. Accordingly, the light extraction efficiency of the light-emitting device **10** may be increased, so that the luminescence efficiency of the light-emitting device **10** may be improved.

(248) Each of the first capping layer and the second capping layer may include a material having a refractive index (at a wavelength of about 589 nm) equal to or greater than about 1.6.

(249) The first capping layer and the second capping layer may each independently be an organic capping layer including an organic material, an inorganic capping layer including an inorganic material, or an organic-inorganic composite capping layer including an organic material and an inorganic material.

(250) At least one of the first capping layer and the second capping layer may each independently include carbocyclic compounds, heterocyclic compounds, amine group-containing compounds, porphyrin derivatives, phthalocyanine derivatives, a naphthalocyanine derivatives, alkali metal complexes, alkaline earth metal complexes, or any combination thereof. The carbocyclic compound, the heterocyclic compound, and the amine group-containing compound may each independently be optionally substituted with a substituent containing O, N, S, Se, Si, F, Cl, Br, I, or any combination thereof.

(251) In an embodiment, at least one of the first capping layer and the second capping layer may each independently include an amine group-containing compound.

(252) In an embodiment, at least one of the first capping layer and the second capping layer may each independently include a compound represented by Formula 201, a compound represented by Formula 202, or any combination thereof.

(253) In embodiments, at least one of the first capping layer and the second capping layer may each independently include one of Compounds HT28 to HT33, one of Compounds CP1 to CP6, β -NPB,

or any combination thereof:

(254) ##STR00149## ##STR00150##

[Film]

(255) The organometallic compound represented by Formula 1 may be included in various films. Accordingly, according to another aspect, a film including the organometallic compound represented by Formula 1 may be provided. The film may be, for example, an optical member (or a light controlling means) (for example, a color filter, a color conversion member, a capping layer, a light extraction efficiency enhancement layer, a selective light absorbing layer, a polarizing layer, or a quantum dot-containing layer), a light-blocking member (for example, a light reflective layer or a light absorbing layer), or a protective member (for example, an insulating layer or a dielectric layer).

(256) [Electronic Apparatus]

(257) The light-emitting device may be included in various electronic apparatuses. In an embodiment, the electronic apparatus including the light-emitting device may be a light-emitting apparatus, an authentication apparatus, or the like.

(258) The electronic apparatus (for example, a light-emitting apparatus) may further include, in addition to the light-emitting device, a color filter, a color conversion layer, or a color filter and a color conversion layer. The color filter and/or the color conversion layer may be located in at least one traveling direction of light emitted from the light-emitting device. In an embodiment, the light emitted from the light-emitting device may be blue light or white light. The light-emitting device may be the same as described above. In an embodiment, the color conversion layer may include quantum dots. The quantum dot may be, for example, a quantum dot as described herein.

(259) The electronic apparatus may include a first substrate. The first substrate may include subpixels, the color filter may include color filter areas respectively corresponding to the subpixels, and the color conversion layer may include color conversion areas respectively corresponding to the subpixels.

(260) A pixel-defining layer may be located among the subpixels to define each of the subpixels.

(261) The color filter may further include color filter areas and light-shielding patterns located among the color filter areas, and the color conversion layer may include color conversion areas and light-shielding patterns located among the color conversion areas.

(262) The color filter areas (or the color conversion areas) may include a first area emitting first-color light, a second area emitting second-color light, and/or a third area emitting third-color light, and the first-color light, the second-color light, and/or the third-color light may have different maximum emission wavelengths from one another. In an embodiment, the first-color light may be red light, the second-color light may be green light, and the third-color light may be blue light. In an embodiment, the color filter areas (or the color conversion areas) may include quantum dots. For example, the first area may include a red quantum dot, the second area may include a green quantum dot, and the third area may not include a quantum dot. The quantum dot may be the same as described in the specification. Each of the first area, the second area, and/or the third area may further include a scatterer.

(263) In an embodiment, the light-emitting device may emit first light, the first area may absorb the first light to emit first first-color light, the second area may absorb the first light to emit second first-color light, and the third area may absorb the first light to emit third first-color light. In this regard, the first first-color light, the second first-color light, and the third first-color light may have different maximum emission wavelengths from one another. For example, the first light may be blue light, the first first-color light may be red light, the second first-color light may be green light, and the third first-color light may be blue light.

(264) The electronic apparatus may further include a thin-film transistor in addition to the light-emitting device as described above. The thin-film transistor may include a source electrode, a drain electrode, and an active layer, wherein any one of the source electrode and the drain electrode may

be electrically connected to any one of the first electrode and the second electrode of the light-emitting device.

(265) The thin-film transistor may further include a gate electrode, a gate insulating film, or the like.

(266) The active layer may include crystalline silicon, amorphous silicon, organic semiconductor, oxide semiconductor, or the like.

(267) The electronic apparatus may further include a sealing portion for sealing the light-emitting device. The sealing portion and/or the color conversion layer may be located between the color filter and the light-emitting device. The sealing portion may allow light from the light-emitting device to be extracted to the outside, and may simultaneously prevent ambient air and/or moisture from penetrating into the light-emitting device. The sealing portion may be a sealing substrate including a transparent glass substrate or a plastic substrate. The sealing portion may be a thin-film encapsulation layer including at least one of an organic layer and/or an inorganic layer. When the sealing portion is a thin film encapsulation layer, the electronic apparatus may be flexible.

(268) Various functional layers may be additionally located on the sealing portion, in addition to the color filter and/or the color conversion layer, according to the use of the electronic apparatus. Examples of the functional layers may include a touch screen layer, a polarizing layer, an authentication apparatus, and the like. The touch screen layer may be a pressure-sensitive touch screen layer, a capacitive touch screen layer, or an infrared touch screen layer. The authentication apparatus may be, for example, a biometric authentication apparatus that authenticates an individual by using biometric information of a living body (for example, fingertips, pupils, etc.).

(269) The authentication apparatus may further include, in addition to the light-emitting device, a biometric information collector.

(270) The electronic apparatus may be applied to various displays, such as light sources, lighting, personal computers (for example, a mobile personal computer), mobile phones, digital cameras, electronic organizers, electronic dictionaries, electronic game machines, medical instruments (for example, electronic thermometers, sphygmomanometers, blood glucose meters, pulse measurement devices, pulse wave measurement devices, electrocardiogram displays, ultrasonic diagnostic devices, or endoscope displays), fish finders, various measuring instruments, meters (for example, meters for a vehicle, an aircraft, and a vessel), projectors, and the like.

(271) [Description of FIGS. 2 and 3]

(272) FIG. 2 is a schematic cross-sectional view of an electronic apparatus according to an embodiment of the disclosure.

(273) The electronic apparatus of FIG. 2 includes a substrate **100**, a thin-film transistor (TFT), a light-emitting device, and an encapsulation portion **300** that seals the light-emitting device.

(274) The substrate **100** may be a flexible substrate, a glass substrate, or a metal substrate. A buffer layer **210** may be located on the substrate **100**. The buffer layer **210** may prevent penetration of impurities through the substrate **100** and may provide a flat surface on the substrate **100**.

(275) A TFT may be located on the buffer layer **210**. The TFT may include an active layer **220**, a gate electrode **240**, a source electrode **260**, and a drain electrode **270**.

(276) The active layer **220** may include an inorganic semiconductor such as silicon or polysilicon, an organic semiconductor, or an oxide semiconductor, and may include a source region, a drain region, and a channel region.

(277) A gate insulating film **230** for insulating the active layer **220** from the gate electrode **240** may be located on the active layer **220**, and the gate electrode **240** may be located on the gate insulating film **230**.

(278) An interlayer insulating film **250** may be located on the gate electrode **240**. The interlayer insulating film **250** may be located between the gate electrode **240** and the source electrode **260** to insulate the gate electrode **240** from the source electrode **260** and between the gate electrode **240** and the drain electrode **270** to insulate the gate electrode **240** from the drain electrode **270**.

(279) The source electrode **260** and the drain electrode **270** may be located on the interlayer insulating film **250**. The interlayer insulating film **250** and the gate insulating film **230** may be formed to expose the source region and the drain region of the active layer **220**, and the source electrode **260** and the drain electrode **270** may be in contact with the exposed portions of the source region and the drain region of the active layer **220**.

(280) The TFT is electrically connected to a light-emitting device to drive the light-emitting device, and is covered by a passivation layer **280**. The passivation layer **280** may include an inorganic insulating film, an organic insulating film, or any combination thereof. A light-emitting device is provided on the passivation layer **280**. The light-emitting device may include a first electrode **110**, an interlayer **130**, and a second electrode **150**.

(281) The first electrode **110** may be located on the passivation layer **280**. The passivation layer **280** may not completely cover the drain electrode **270** and may expose a portion of the drain electrode **270**, and the first electrode **110** may be electrically connected to the exposed portion of the drain electrode **270**.

(282) A pixel defining layer **290** containing an insulating material may be located on the first electrode **110**. The pixel defining layer **290** may expose a portion of the first electrode **110**, and the interlayer **130** may be formed in the exposed portion of the first electrode **110**. The pixel defining layer **290** may be a polyimide or polyacrylic organic film. Although not shown in FIG. 2, at least some layers of the interlayer **130** may extend beyond the upper portion of the pixel defining layer **290** to be provided in the form of a common layer.

(283) The second electrode **150** may be disposed on the interlayer **130**, and a capping layer **170** may be additionally formed on the second electrode **150**. The capping layer **170** may be formed to cover the second electrode **150**.

(284) The encapsulation portion **300** may be located on the capping layer **170**. The encapsulation portion **300** may be located on a light-emitting device to protect the light-emitting device from moisture and/or oxygen. The encapsulation portion **300** may include: an inorganic film including silicon nitride (SiN.sub.x), silicon oxide (SiO.sub.X), indium tin oxide, indium zinc oxide, or any combination thereof; an organic film including polyethylene terephthalate, polyethylene naphthalate, polycarbonate, polyimide, polyethylene sulfonate, polyoxymethylene, polyarylate, hexamethyldisiloxane, an acrylic resin (for example, polymethyl methacrylate, polyacrylic acid, or the like), an epoxy-based resin (for example, aliphatic glycidyl ether (AGE), or the like), or any combination thereof; or any combination of the inorganic film and the organic film.

(285) FIG. 3 is a schematic cross-sectional view of an electronic apparatus according to an embodiment of the disclosure.

(286) The electronic apparatus of FIG. 3 is the same as the electronic apparatus of FIG. 2, except that a light-shielding pattern **500** and a functional region **400** are additionally located on the encapsulation portion **300**. The functional region **400** may be a color filter area, a color conversion area, or a combination of the color filter area and the color conversion area. In an embodiment, the light-emitting device included in the electronic apparatus of FIG. 3 may be a tandem light-emitting device.

(287) [Manufacture Method]

(288) Respective layers included in the hole transport region, the emission layer, and respective layers included in the electron transport region may be formed in a certain region by using one or more suitable methods selected from vacuum deposition, spin coating, casting, Langmuir-Blodgett (LB) deposition, ink-jet printing, laser-printing, and laser-induced thermal imaging.

(289) When respective layers included in the hole transport region, the emission layer, and respective layers included in the electron transport region are formed by vacuum deposition, the deposition may be performed at a deposition temperature of about 100° C. to about 500° C., a vacuum degree of about 10.sup.-8 torr to about 10.sup.-3 torr, and a deposition speed of about 0.01 Å/sec to about 100 Å/sec, depending on a material to be included in a layer to be formed and

the structure of a layer to be formed.

Definitions of Terms

(290) The term “C.sub.3-C.sub.60 carbocyclic group” as used herein may be a cyclic group consisting only of carbon as a ring-forming atom and having 3 to 60 carbon atoms, and the term “C.sub.1-C.sub.60 heterocyclic group” as used herein may be a cyclic group that has 1 to 60 carbon atoms and further has, in addition to carbon, at least one heteroatom as a ring-forming atom. The C.sub.3-C.sub.60 carbocyclic group and the C.sub.1-C.sub.60 heterocyclic group may each be a monocyclic group consisting of one ring or a polycyclic group in which two or more rings are condensed with each other. In an embodiment, a C.sub.1-C.sub.60 heterocyclic group may have 3 to 61 ring-forming atoms.

(291) The term “cyclic group” as used herein may include the C.sub.3-C.sub.60 carbocyclic group and the C.sub.1-C.sub.60 heterocyclic group.

(292) The term “ π electron-rich C.sub.3-C.sub.60 cyclic group” as used herein may be a cyclic group that has three to sixty carbon atoms and may not include —N= as a ring-forming moiety, and the term “ π electron-deficient nitrogen-containing C.sub.1-C.sub.60 cyclic group” as used herein may be a heterocyclic group that has one to sixty carbon atoms and may include —N= as a ring-forming moiety.

(293) In an embodiment, the C.sub.3-C.sub.60 carbocyclic group may be a T1 group or a condensed cyclic group in which two or more T1 groups are condensed with each other (for example, a cyclopentadiene group, an adamantane group, a norbornane group, a benzene group, a pentalene group, a naphthalene group, an azulene group, an indacene group, an acenaphthylene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a perylene group, a pentaphene group, a heptalene group, a naphthacene group, a picene group, a hexacene group, a pentacene group, a rubicene group, a coronene group, an ovalene group, an indene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, an indenophenanthrene group, or an indenoanthracene group), the C.sub.1-C.sub.60 heterocyclic group may be a T2 group T2, a condensed cyclic group in which two or more T2 groups are condensed with each other, or a condensed cyclic group in which at least one T2 group and at least one T1 group are condensed with each other (for example, a pyrrole group, a thiophene group, a furan group, an indole group, a benzoindole group, a naphthoindole group, an isoindole group, a benzoisoindole group, a naphthoisoindole group, a benzosilole group, a benzothiophene group, a benzofuran group, a carbazole group, a dibenzosilole group, a dibenzothiophene group, a dibenzofuran group, an indenocarbazole group, an indolocarbazole group, a benzofurocarbazole group, a benzothienocarbazole group, a benzosilolocarbazole group, a benzoindolocarbazole group, a benzocarbazole group, a benzonaphthofuran group, a benzonaphthothiophene group, a benzonaphthosilole group, a benzofurodibenzofuran group, a benzofurodibenzothiophene group, a benzothienodibenzothiophene group, a pyrazole group, an imidazole group, a triazole group, an oxazole group, an isoxazole group, an oxadiazole group, a thiazole group, an isothiazole group, a thiadiazole group, a benzopyrazole group, a benzimidazole group, a benzoxazole group, a benzoisoxazole group, a benzothiazole group, a benzoisothiazole group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a benzoisoquinoline group, a quinoxaline group, a benzoquinoxaline group, a quinazoline group, a benzoquinazoline group, a phenanthroline group, a cinnoline group, a phthalazine group, a naphthyridine group, an imidazopyridine group, an imidazopyrimidine group, an imidazotriazine group, an imidazopyrazine group, an imidazopyridazine group, an azacarbazole group, an azafluorene group, an azadibenzosilole group, an azadibenzothiophene group, an azadibenzofuran group, etc.), the π electron-rich C.sub.3-C.sub.60 cyclic group may be a T1 group, a condensed cyclic group in which two or more T1 groups are condensed with each other, a T3 group T3, a condensed cyclic group in which two or

more T3 groups are condensed with each other, or a condensed cyclic group in which at least one T3 group and at least one T1 group are condensed with each other (for example, the C.sub.3-C.sub.60 carbocyclic group, a 1H-pyrrole group, a silole group, a borole group, a 2H-pyrrole group, a 3H-pyrrole group, a thiophene group, a furan group, an indole group, a benzoindole group, a naphthoindole group, an isoindole group, a benzoisoindole group, a naphthoisoindole group, a benzosilole group, a benzothiophene group, a benzofuran group, a carbazole group, a dibenzosilole group, a dibenzothiophene group, a dibenzofuran group, an indenocarbazole group, an indolocarbazole group, a benzofurocarbazole group, a benzothienocarbazole group, a benzosilolocarbazole group, a benzoindolocarbazole group, a benzocarbazole group, a benzonaphthofuran group, a benzonaphthothiophene group, a benzonaphthosilole group, a benzofurodibenzofuran group, a benzofurodibenzothiophene group, a benzothienodibenzothiophene group, etc.), the π electron-deficient nitrogen-containing C.sub.1-C.sub.60 cyclic group may be a T4 group, a condensed cyclic group in which two or more T4 groups are condensed with each other, a condensed cyclic group in which at least one T4 group and at least one T1 group are condensed with each other, a condensed cyclic group in which at least one T4 group and at least one T3 group are condensed with each other, or a condensed cyclic group in which at least one T4 group, at least one T1 group, and at least one T3 group are condensed with one another (for example, a pyrazole group, an imidazole group, a triazole group, an oxazole group, an isoxazole group, an oxadiazole group, a thiazole group, an isothiazole group, a thiadiazole group, a benzopyrazole group, a benzimidazole group, a benzoxazole group, a benzoisoxazole group, a benzothiazole group, a benzoisothiazole group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a benzoisoquinoline group, a quinoxaline group, a benzoquinoxaline group, a quinazoline group, a benzoquinazoline group, a phenanthroline group, a cinnoline group, a phthalazine group, a naphthyridine group, an imidazopyridine group, an imidazopyrimidine group, an imidazotriazine group, an imidazopyrazine group, an imidazopyridazine group, an azacarbazole group, an azafluorene group, an azadibenzosilole group, an azadibenzothiophene group, an azadibenzofuran group, etc.), wherein the T1 group may be a cyclopropane group, a cyclobutane group, a cyclopentane group, a cyclohexane group, a cycloheptane group, a cyclooctane group, a cyclobutene group, a cyclopentene group, a cyclopentadiene group, a cyclohexene group, a cyclohexadiene group, a cycloheptene group, an adamantane group, a norbornane (or a bicyclo[2.2.1]heptane) group, a norbornene group, a bicyclo[1.1.1]pentane group, a bicyclo[2.1.1]hexane group, a bicyclo[2.2.2]octane group, or a benzene group, the T2 group may be a furan group, a thiophene group, a 1H-pyrrole group, a silole group, a borole group, a 2H-pyrrole group, a 3H-pyrrole group, an imidazole group, a pyrazole group, a triazole group, a tetrazole group, an oxazole group, an isoxazole group, an oxadiazole group, a thiazole group, an isothiazole group, a thiadiazole group, an azasilole group, an azaborole group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a tetrazine group, a pyrrolidine group, an imidazolidine group, a dihydropyrrole group, a piperidine group, a tetrahydropyridine group, a dihydropyridine group, a hexahydropyrimidine group, a tetrahydropyrimidine group, a dihydropyrimidine group, a piperazine group, a tetrahydropyrazine group, a dihydropyrazine group, a tetrahydropyridazine group, or a dihydropyridazine group, the T3 group may be a furan group, a thiophene group, a 1H-pyrrole group, a silole group, or a borole group, and the T4 group may be a 2H-pyrrole group, a 3H-pyrrole group, an imidazole group, a pyrazole group, a triazole group, a tetrazole group, an oxazole group, an isoxazole group, an oxadiazole group, a thiazole group, an isothiazole group, a thiadiazole group, an azasilole group, an azaborole group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, or a tetrazine group.

(294) The terms “cyclic group”, “C.sub.3-C.sub.60 carbocyclic group”, “C.sub.1-C.sub.60 heterocyclic group”, “ π electron-rich C.sub.3-C.sub.60 cyclic group”, or “ π electron-deficient

nitrogen-containing C.sub.1-C.sub.60 cyclic group” as used herein may be a group condensed to any cyclic group or a polyvalent group (for example, a divalent group, a trivalent group, a tetravalent group, etc.), depending on the structure of a formula in connection with which the terms are used. In an embodiment, “a benzene group” may be a benzo group, a phenyl group, a phenylene group, or the like, which may be easily understood by one of ordinary skill in the art according to the structure of a formula including the “benzene group.”

(295) Examples of the monovalent C.sub.3-C.sub.60 carbocyclic group and the monovalent C.sub.1-C.sub.60 heterocyclic group may include a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, and a monovalent non-aromatic condensed heteropolycyclic group, and examples of the divalent C.sub.3-C.sub.60 carbocyclic group and the monovalent C.sub.1-C.sub.60 heterocyclic group may include a C.sub.3-C.sub.10 cycloalkylene group, a C.sub.1-C.sub.10 heterocycloalkylene group, a C.sub.3-C.sub.10 cycloalkenylene group, a C.sub.1-C.sub.10 heterocycloalkenylene group, a C.sub.6-C.sub.60 arylene group, a C.sub.1-C.sub.60 heteroarylene group, a divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group.

(296) The term “C.sub.1-C.sub.60 alkyl group” as used herein may be a linear or branched aliphatic hydrocarbon monovalent group that has 1 to 60 carbon atoms, and examples thereof may include a methyl group, an ethyl group, an n-propyl group, an isopropyl group, an n-butyl group, a sec-butyl group, an isobutyl group, a tert-butyl group, an n-pentyl group, a tert-pentyl group, a neopentyl group, an isopentyl group, a sec-pentyl group, a 3-pentyl group, a sec-isopentyl group, an n-hexyl group, an isohexyl group, a sec-hexyl group, a tert-hexyl group, an n-heptyl group, an isoheptyl group, a sec-heptyl group, a tert-heptyl group, an n-octyl group, an isooctyl group, a sec-octyl group, a tert-octyl group, an n-nonyl group, an isononyl group, a sec-nonyl group, a tert-nonyl group, an n-decyl group, an isodecyl group, a sec-decyl group, and a tert-decyl group. The term “C.sub.1-C.sub.60 alkylene group” as used herein may be a divalent group having a same structure as the C.sub.1-C.sub.60 alkyl group.

(297) The term “C.sub.2-C.sub.60 alkenyl group” as used herein may be a monovalent hydrocarbon group having at least one carbon-carbon double bond in the middle or at the terminus of the C.sub.2-C.sub.60 alkyl group, and examples thereof may include an ethenyl group, a propenyl group, and a butenyl group. The term “C.sub.2-C.sub.60 alkenylene group” as used herein may be a divalent group having a same structure as the C.sub.2-C.sub.60 alkenyl group.

(298) The term “C.sub.2-C.sub.60 alkynyl group” as used herein may be a monovalent hydrocarbon group having at least one carbon-carbon triple bond in the middle or at the terminus of the C.sub.2-C.sub.60 alkyl group, and examples thereof may include an ethynyl group and a propynyl group. The term “C.sub.2-C.sub.60 alkynylene group” as used herein may be a divalent group having a same structure as the C.sub.2-C.sub.60 alkynyl group.

(299) The term “C.sub.1-C.sub.60 alkoxy group” as used herein may be a monovalent group represented by —O(A.sub.101) (wherein A.sub.101 is a C.sub.1-C.sub.60 alkyl group), and examples thereof may include a methoxy group, an ethoxy group, and an isopropoxy group.

(300) The term “C.sub.3-C.sub.10 cycloalkyl group” as used herein may be a monovalent saturated hydrocarbon cyclic group having 3 to 10 carbon atoms, and examples thereof may include a cyclopropyl group, a cyclobutyl group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group (or bicyclo[2.2.1]heptyl group), a bicyclo[1.1.1]pentyl group, a bicyclo[2.1.1]hexyl group, and a bicyclo[2.2.2]octyl group. The term “C.sub.3-C.sub.10 cycloalkylene group” as used herein may be a divalent group having a same structure as the C.sub.3-C.sub.10 cycloalkyl group.

(301) The term “C.sub.1-C.sub.10 heterocycloalkyl group” as used herein may be a monovalent cyclic group that further includes, in addition to a carbon atom, at least one heteroatom as a ring-

forming atom and has 1 to 10 carbon atoms, and examples thereof may include a 1,2,3,4-oxatriazolidinyl group, a tetrahydrofuranyl group, and a tetrahydrothiophenyl group. The term “C.sub.1-C.sub.10 heterocycloalkylene group” as used herein may be a divalent group having a same structure as the C.sub.1-C.sub.10 heterocycloalkyl group.

(302) The term “C.sub.3-C.sub.10 cycloalkenyl group” as used herein may be a monovalent cyclic group that has 3 to 10 carbon atoms and at least one carbon-carbon double bond in the ring thereof and no aromaticity, and examples thereof may include a cyclopentenyl group, a cyclohexenyl group, and a cycloheptenyl group. The term “C.sub.3-C.sub.10 cycloalkenylene group” as used herein may be a divalent group having a same structure as the C.sub.3-C.sub.10 cycloalkenyl group.

(303) The term “C.sub.1-C.sub.10 heterocycloalkenyl group” as used herein may be a monovalent cyclic group that has, in addition to a carbon atom, at least one heteroatom as a ring-forming atom, 1 to 10 carbon atoms, and at least one double bond in the cyclic structure thereof. Examples of the C.sub.1-C.sub.10 heterocycloalkenyl group may include a 4,5-dihydro-1,2,3,4-oxatriazolyl group, a 2,3-dihydrofuranyl group, and a 2,3-dihydrothiophenyl group. The term “C.sub.1-C.sub.10 heterocycloalkenylene group” as used herein may be a divalent group having a same structure as the C.sub.1-C.sub.10 heterocycloalkenyl group.

(304) The term “C.sub.6-C.sub.60 aryl group” as used herein may be a monovalent group having a carbocyclic aromatic system having 6 to 60 carbon atoms, and the term “C.sub.6-C.sub.60 arylene group” as used herein may be a divalent group having a carbocyclic aromatic system having 6 to 60 carbon atoms. Examples of the C.sub.6-C.sub.60 aryl group may include a phenyl group, a pentalenyl group, a naphthyl group, an azulenyl group, an indacenyl group, an acenaphthyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a heptalenyl group, a naphthacenyl group, a picenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, and an ovalenyl group. When the C.sub.6-C.sub.60 aryl group and the C.sub.6-C.sub.60 arylene group each include two or more rings, the rings may be condensed with each other.

(305) The term “C.sub.1-C.sub.60 heteroaryl group” as used herein may be a monovalent group having a heterocyclic aromatic system that has, in addition to a carbon atom, at least one heteroatom as a ring-forming atom, and 1 to 60 carbon atoms. The term “C.sub.1-C.sub.60 heteroarylene group” as used herein may be a divalent group having a heterocyclic aromatic system that has, in addition to a carbon atom, at least one heteroatom as a ring-forming atom, and 1 to 60 carbon atoms. Examples of the C.sub.1-C.sub.60 heteroaryl group may include a pyridinyl group, a pyrimidinyl group, a pyrazinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, a benzoquinolinyl group, an isoquinolinyl group, a benzoisoquinolinyl group, a quinoxalinyl group, a benzoquinoxalinyl group, a quinazolinyl group, a benzoquinazolinyl group, a cinnolinyl group, a phenanthrolinyl group, a phthalazinyl group, and a naphthyridinyl group. When the C.sub.1-C.sub.60 heteroaryl group and the C.sub.1-C.sub.60 heteroarylene group each include two or more rings, the rings may be condensed with each other.

(306) The term “monovalent non-aromatic condensed polycyclic group” as used herein may be a monovalent group (for example, having 8 to 60 carbon atoms) having two or more rings condensed to each other, only carbon atoms as ring-forming atoms, and no aromaticity in its entire molecular structure. Examples of the monovalent non-aromatic condensed polycyclic group may include an indenyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, an indenophenanthrenyl group, and an indeno anthracenyl group. The term “divalent non-aromatic condensed polycyclic group” as used herein may be a divalent group having a same structure as a monovalent non-aromatic condensed polycyclic group.

(307) The term “monovalent non-aromatic condensed heteropolycyclic group” as used herein may be a monovalent group (for example, having 1 to 60 carbon atoms) having two or more rings

condensed to each other, at least one heteroatom other than carbon atoms, as a ring-forming atom, and non-aromaticity in its entire molecular structure. Examples of the monovalent non-aromatic condensed heteropolycyclic group may include a pyrrolyl group, a thiophenyl group, a furanyl group, an indolyl group, a benzoindolyl group, a naphthoindolyl group, an isoindolyl group, a benzoisoindolyl group, a naphthoisoindolyl group, a benzosilolyl group, a benzothiophenyl group, a benzofuranyl group, a carbazolyl group, a dibenzosilolyl group, a dibenzothiophenyl group, a dibenzofuranyl group, an azacarbazolyl group, an azafluorenyl group, an azadibenzosilolyl group, an azadibenzothiophenyl group, an azadibenzofuranyl group, a pyrazolyl group, an imidazolyl group, a triazolyl group, a tetrazolyl group, an oxazolyl group, an isoxazolyl group, a thiazolyl group, an isothiazolyl group, an oxadiazolyl group, a thiadiazolyl group, a benzopyrazolyl group, a benzimidazolyl group, a benzoxazolyl group, a benzothiazolyl group, a benzoxadiazolyl group, a benzothiadiazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an imidazotriazinyl group, an imidazopyrazinyl group, an imidazopyridazinyl group, an indeno carbazolyl group, an indolocarbazolyl group, a benzofurocarbazolyl group, a benzothienocarbazolyl group, a benzosilolocarbazolyl group, a benzoindolocarbazolyl group, a benzocarbazolyl group, a benzonaphthofuranyl group, a benzonaphthothiophenyl group, a benzonaphtho silolyl group, a benzofurodibenzofuranyl group, a benzofurodibenzothiophenyl group, and a benzothienodibenzothiophenyl group. The term “divalent non-aromatic condensed heteropolycyclic group” as used herein may be a divalent group having a same structure as a monovalent non-aromatic condensed heteropolycyclic group.

(308) The term “C.sub.6-C.sub.60 aryloxy group” as used herein may be represented by —O(A.sub.102) (wherein A.sub.102 is a C.sub.6-C.sub.60 aryl group), and the term “C.sub.6-C.sub.60 arylthio group” as used herein may be represented by —S(A.sub.103) (wherein A.sub.103 is a C.sub.6-C.sub.60 aryl group).

(309) The term “C.sub.7-C.sub.60 aryl alkyl group” as used herein may be represented by -(A.sub.104)(A.sub.105) (where A.sub.104 may be a C.sub.1-C.sub.54 alkylene group, and A105 may be a C.sub.6-C.sub.59 aryl group), and the term “C.sub.2-C.sub.60 heteroaryl alkyl group” as used herein may be represented by -(A.sub.106)(A.sub.107) (where A106 may be a C.sub.1-C.sub.59 alkylene group, and A107 may be a C.sub.1-C.sub.59 heteroaryl group).

(310) The term “R.sub.10a” as used herein may be: deuterium (-D), —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, or a nitro group; a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, or a C.sub.1-C.sub.60 alkoxy group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryloxy group, a C.sub.1-C.sub.60 heteroarylthio group, —Si(Q.sub.11)(Q.sub.12)(Q.sub.13), —N(Q.sub.11)(Q.sub.12), —B(Q.sub.11)(Q.sub.12), —C(=O)(Q.sub.11), —S(=O).sub.2(Q.sub.11), —P(=O)(Q.sub.11)(Q.sub.12), or any combination thereof; a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryloxy group, or a C.sub.1-C.sub.60 heteroarylthio group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryloxy group, a C.sub.1-C.sub.60 heteroarylthio group, —Si(Q.sub.21)(Q.sub.22)(Q.sub.23), —N(Q.sub.21)(Q.sub.22), —B(Q.sub.21)(Q.sub.22), —C(=O)(Q.sub.21), —S(=O).sub.2(Q.sub.21), —P(=O)(Q.sub.21)(Q.sub.22), or any combination thereof; or —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), or —P(=O)(Q.sub.31)(Q.sub.32).

(311) Q.sub.11 to Q.sub.13, Q.sub.21 to Q.sub.23, and Q.sub.31 to Q.sub.33 as used herein may each independently be: hydrogen; deuterium; —F; —Cl; —Br; —I; a hydroxyl group; a cyano group; a nitro group; a C.sub.1-C.sub.60 alkyl group; a C.sub.2-C.sub.60 alkenyl group; a C.sub.2-C.sub.60 alkynyl group; a C.sub.1-C.sub.60 alkoxy group; a C.sub.3-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, each unsubstituted or substituted with deuterium, —F, a cyano group, a C.sub.1-C.sub.60 alkyl group, a C.sub.1-C.sub.6 alkoxy group, a phenyl group, a biphenyl group, or any combination thereof.

(312) The term “heteroatom” as used herein may be any atom other than a carbon atom or a hydrogen atom. Examples of the heteroatom may include O, S, N, P, Si, B, Ge, Se, or any combination thereof.

(313) The term “the third-row transition metal” as used herein may include hafnium (Hf), tantalum (Ta), tungsten (W), rhenium (Re), osmium (Os), iridium (Ir), platinum (Pt), gold (Au), etc.

(314) The term “Ph” as used herein refers to a phenyl group, the term “Me” as used herein refers to a methyl group, the term “Et” as used herein refers to an ethyl group, the term “ter-Bu” or “Bu.sup.t” as used herein refers to a tert-butyl group, and the term “OMe” as used herein refers to a methoxy group.

(315) The term “biphenyl group” as used herein may be “a phenyl group substituted with a phenyl group.” For example, the “biphenyl group” may be a substituted phenyl group having a C.sub.6-C.sub.60 aryl group as a substituent.

(316) The term “terphenyl group” as used herein may be “a phenyl group substituted with a biphenyl group”. For example, the “terphenyl group” may be a substituted phenyl group having, as a substituent, a C.sub.6-C.sub.60 aryl group substituted with a C.sub.6-C.sub.60 aryl group.

(317) In the specification, *, *', and *'' as used herein, unless defined otherwise, each represent a binding site to a neighboring atom in a corresponding formula or moiety.

(318) Hereinafter, a compound according to embodiments and a light-emitting device according to embodiments will be described in detail with reference to Examples. The wording “B was used instead of A” used in describing Examples refers to that an identical molar equivalent of B was used in place of A.

EXAMPLES

Synthesis Example 1: Synthesis of Compound BD2

(319) ##STR00151## ##STR00152## ##STR00153##

(1) Synthesis of Intermediate [1-1]

(320) 21.0 g (62 mmol) of benzimidazole, 20.2 g (74 mmol) of 3-bromoanisole, 26.4 g (124 mmol) of potassium phosphate tribasic, 1,180 mg (6.2 mmol) of iodo copper, and 760 mg (6.2 mmol) of picolinic acid were added to a reaction vessel, and suspended in 650 mL of dimethyl sulfoxide. The reaction mixture was heated, and stirred at 160° C. for 12 hours. After completion of the reaction, the reaction mixture was cooled to room temperature, and extracted with ethyl acetate. An organic layer extracted therefrom was washed with a saturated aqueous sodium chloride solution, and dried using sodium sulfate. A residue obtained by removing the solvent therefrom was separated by column chromatography to obtain 21.2 g (40 mmol) of Intermediate [1-1].

(321) (2) Synthesis of Intermediate [1-2]

(322) 21.2 g (40 mmol) of Intermediate [1-1] was added to a reaction vessel, and suspended in 400 mL of HCl/AcOH (volume ratio of 4:1). The reaction temperature was raised to 120° C., and the reaction mixture was stirred for 12 hours. After completion of the reaction, the reaction mixture was cooled to room temperature, and extracted with ethyl acetate. An organic layer extracted therefrom was washed with a saturated aqueous sodium chloride solution, and dried using sodium sulfate. A residue obtained by removing the solvent therefrom was separated by column chromatography to obtain 15.3 g (32 mmol) of Intermediate [1-2].

(323) (3) Synthesis of Intermediate [1-3]

(324) 15.3 g (32 mmol) of Intermediate [1-2], 12.1 g (35 mmol) of 1,3-dibromobenzene, 18.5 g (70

mmol) of potassium phosphate tribasic, 560 mg (3.2 mmol) of iodo copper, and 420 mg (3.5 mmol) of picolinic acid were added to a reaction vessel, and suspended in 400 mL of dimethyl sulfoxide. The reaction mixture was heated, and stirred at 160° C. for 12 hours. After completion of the reaction, the reaction mixture was cooled to room temperature, and extracted with ethyl acetate. An organic layer extracted therefrom was washed with a saturated aqueous sodium chloride solution, and dried using sodium sulfate. A residue obtained by removing the solvent therefrom was separated by column chromatography to obtain 14.5 g (28 mmol) of Intermediate [1-3].

(325) (4) Synthesis of Intermediate [1-4]

(326) 14.5 g (28 mmol) of Intermediate [1-3], 3.0 g (28 mmol) of benzene-1,2-diamine, 460 mg (0.5 mmol) of tris(dibenzylideneacetone)dipalladium, 480 mg (1.0 mmol) of Xphos, and 5.0 g (52 mmol) of sodium tert-butoxide were added to a reaction vessel, and suspended in 260 mL of toluene. The reaction temperature was raised to 110° C., and the reaction mixture was stirred for 6 hours. After completion of the reaction, the reaction mixture was cooled to room temperature, and extracted with ethyl acetate. An organic layer extracted therefrom was washed with a saturated aqueous sodium chloride solution, and dried using sodium sulfate. A residue obtained by removing the solvent therefrom was separated by column chromatography to obtain 14.2 g (19 mmol) of Intermediate [1-4].

(327) (5) Synthesis of Intermediate [1-5]

(328) 14.2 g (19 mmol) of Intermediate [1-4], 2.1 g (9 mmol) of 1,4-dibromobenzene, 430 mg (0.4 mmol) of tris(dibenzylideneacetone)dipalladium, 450 mg (0.8 mmol) of XPhos, and 4.5 g (40 mmol) of sodium tert-butoxide were added to a reaction vessel, and suspended in 250 mL of toluene. The reaction temperature was raised to 110° C., and the reaction mixture was stirred for 6 hours. After completion of the reaction, the reaction mixture was cooled to room temperature, and extracted with ethyl acetate. An organic layer extracted therefrom was washed with a saturated aqueous sodium chloride solution, and dried using sodium sulfate. A residue obtained by removing the solvent therefrom was separated by column chromatography to obtain 3.1 g (5 mmol) of Intermediate [1-5].

(329) (6) Synthesis of Intermediate [1-6]

(330) 3.1 g (5 mmol) of Intermediate [1-5], 20 mL (250 mmol) of triethyl orthoformate, and 3.5 g (35 mmol) of a HCl 35 wt % solution were added to a reaction vessel, heated, and stirred at 80° C. for 12 hours. After completion of the reaction, the reaction mixture was cooled to room temperature, and a residue obtained by removing the solvent therefrom was separated by column chromatography to obtain 3.2 g (4 mmol) of Intermediate [1-6].

(331) (7) Synthesis of Intermediates [1-7] and [1-8]

(332) 3.2 g (4 mmol) of Intermediate [1-6] and 0.6 g (16 mmol) of iodomethane were added to a reaction vessel, and suspended in 110 mL of dichloromethane. The reaction temperature was raised to 50° C., and the reaction mixture was stirred for 12 hours. After completion of the reaction, the reaction mixture was cooled to room temperature, and extracted with ethyl acetate. An organic layer extracted therefrom was washed with a saturated aqueous sodium chloride solution, and dried using sodium sulfate. A residue obtained by removing the solvent therefrom was separated by column chromatography to obtain 3.8 g (4 mmol) of Intermediate [1-7]. Immediately afterwards, 2.3 g (14 mmol) of ammonium hexafluorophosphate was added to a reaction vessel, and suspended in a solution of methanol and water at a ratio of 2:1. The reaction mixture was stirred at room temperature for 12 hours. The resulting solid was filtered, and separated by column chromatography to obtain 3.5 g (3 mmol) of Intermediate [1-8].

(333) (8) Synthesis of Compound BD2

(334) 3.5 g (3 mmol) of Intermediate [1-8], 1.0 g (4 mmol) of dichloro(1,5-cyclooctadiene)platinum, and 0.4 g (6 mmol) of sodium acetate were suspended in 100 mL of dioxane. The reaction mixture was heated, and stirred at 110° C. for 72 hours. After completion of the reaction, the reaction mixture was cooled to room temperature, and extracted with ethyl acetate.

An organic layer extracted therefrom was washed with a saturated aqueous sodium chloride solution, and dried using sodium sulfate. A residue obtained by removing the solvent therefrom was separated by column chromatography to obtain 0.7 g (1 mmol) of Compound BD2.

Synthesis Example 2: Synthesis of Compound BD9

(335) ##STR00154## ##STR00155## ##STR00156##

(1) Synthesis of Intermediate [2-1]

(336) 21.0 g (60 mmol) of 3,4-dimethyl benzimidazole, 20.2 g (72 mmol) 3-bromoanisole, 26.4 g (124 mmol) of potassium phosphate tribasic, 1,180 mg (6.2 mmol) of iodo copper, and 760 mg (6.2 mmol) of picolinic acid were added to a reaction vessel, and suspended in 650 mL of dimethyl sulfoxide. The reaction mixture was heated, and stirred at 160° C. for 12 hours. After completion of the reaction, the reaction mixture was cooled to room temperature, and extracted with ethyl acetate. An organic layer extracted therefrom was washed with a saturated aqueous sodium chloride solution, and dried using sodium sulfate. A residue obtained by removing the solvent therefrom was separated by column chromatography to obtain 22.3 g (42 mmol) of Intermediate [2-1].

(337) (2) Synthesis of Intermediate [2-2]

(338) 22.3 g (42 mmol) of Intermediate [2-1] was added to a reaction vessel, and suspended in 400 mL of HCl/AcOH (volume ratio of 4:1). The reaction temperature was raised to 120° C., and the reaction mixture was stirred for 12 hours. After completion of the reaction, the reaction mixture was cooled to room temperature, and extracted with ethyl acetate. An organic layer extracted therefrom was washed with a saturated aqueous sodium chloride solution, and dried using sodium sulfate. A residue obtained by removing the solvent therefrom was separated by column chromatography to obtain 14.7 g (32 mmol) of Intermediate [2-2].

(339) (3) Synthesis of Intermediate [2-3]

(340) 14.7 g (32 mmol) of Intermediate [2-2], 12.1 g (35 mmol) of 1,3-dibromobenzene, 18.5 g (70 mmol) of potassium phosphate tribasic, 560 mg (3.2 mmol) of iodo copper, and 420 mg (3.5 mmol) of picolinic acid were added to a reaction vessel, and suspended in 400 mL of dimethyl sulfoxide. The reaction mixture was heated, and stirred at 160° C. for 12 hours. After completion of the reaction, the reaction mixture was cooled to room temperature, and extracted with ethyl acetate. An organic layer extracted therefrom was washed with a saturated aqueous sodium chloride solution, and dried using sodium sulfate. A residue obtained by removing the solvent therefrom was separated by column chromatography to obtain 14.1 g (25 mmol) of Intermediate [2-3].

(341) (4) Synthesis of Intermediate [2-4]

(342) 14.1 g (25 mmol) of Intermediate [2-3], 3.0 g (28 mmol) of benzene-1,2-diamine, 460 mg (0.5 mmol) of tris(dibenzylideneacetone)dipalladium, 480 mg (1.0 mmol) of Xphos, and 5.0 g (52 mmol) of sodium tert-butoxide were added to a reaction vessel, and suspended in 260 mL of toluene. The reaction temperature was raised to 110° C., and the reaction mixture was stirred for 6 hours. After completion of the reaction, the reaction mixture was cooled to room temperature, and extracted with ethyl acetate. An organic layer extracted therefrom was washed with a saturated aqueous sodium chloride solution, and dried using sodium sulfate. A residue obtained by removing the solvent therefrom was separated by column chromatography to obtain 14.0 g (18 mmol) of Intermediate [2-4].

(343) (5) Synthesis of Intermediate [2-5]

(344) 14.0 g (18 mmol) of Intermediate [2-4], 2.1 g (9 mmol) of 1,4-dibromobenzene, 430 mg (0.4 mmol) of tris(dibenzylideneacetone)dipalladium, 450 mg (0.8 mmol) of XPhos, and 4.5 g (40 mmol) of sodium tert-butoxide were added to a reaction vessel, and suspended in 250 mL of toluene. The reaction temperature was raised to 110° C., and the reaction mixture was stirred for 6 hours. After completion of the reaction, the reaction mixture was cooled to room temperature, and extracted with ethyl acetate. An organic layer extracted therefrom was washed with a saturated aqueous sodium chloride solution, and dried using sodium sulfate. A residue obtained by removing the solvent therefrom was separated by column chromatography to obtain 3.0 g (4 mmol) of

Intermediate [2-5].

(345) (6) Synthesis of Intermediate [2-6]

(346) 3.0 g (4 mmol) of Intermediate [2-5], 20 mL (250 mmol) of triethyl orthoformate, and 3.5 g (35 mmol) of a HCl 35 wt % solution were added to a reaction vessel, heated, and stirred at 80° C. for 12 hours. After completion of the reaction, the reaction mixture was cooled to room temperature, and a residue obtained by removing the solvent therefrom was separated by column chromatography to obtain 2.9 g (3 mmol) of Intermediate [2-6].

(347) (7) Synthesis of Intermediates [2-7] and [2-8]

(348) 2.9 g (3 mmol) of Intermediate [2-6] and 0.6 g (16 mmol) of iodomethane were added to a reaction vessel, and suspended in 110 mL of dichloromethane. The reaction temperature was raised to 50° C., and the reaction mixture was stirred for 12 hours. After completion of the reaction, the reaction mixture was cooled to room temperature, and extracted with ethyl acetate. An organic layer extracted therefrom was washed with a saturated aqueous sodium chloride solution, and dried using sodium sulfate. A residue obtained by removing the solvent therefrom was separated by column chromatography to obtain 3.8 g (4 mmol) of Intermediate [2-7]. Immediately afterwards, 2.3 g (14 mmol) of ammonium hexafluorophosphate was added to a reaction vessel, and suspended in a solution of methanol and water at a ratio of 2:1. The reaction mixture was stirred at room temperature for 12 hours. The resulting solid was filtered, and separated by column chromatography to obtain 3.6 g (3 mmol) of Intermediate [2-8].

(349) (8) Synthesis of Compound BD9

(350) 3.6 g (3 mmol) of Intermediate [2-8], 1.0 g (4 mmol) of dichloro(1,5-cyclooctadiene)platinum, and 0.4 g (6 mmol) of sodium acetate were suspended in 100 mL of dioxane. The reaction mixture was heated, and stirred at 110° C. for 72 hours. After completion of the reaction, the reaction mixture was cooled to room temperature, and extracted with ethyl acetate. An organic layer extracted therefrom was washed with a saturated aqueous sodium chloride solution, and dried using sodium sulfate. A residue obtained by removing the solvent therefrom was separated by column chromatography to obtain 0.6 g (1 mmol) of Compound BD9.

Synthesis Example 3: Synthesis of Compounds BD1 to BD104

(351) Compounds BD1 to BD104 were synthesized in substantially the same manner as Compound BD2, except that a different starting material from that of Compound BD2 was used.

(352) ¹H NMR and MS/FAB of the compounds synthesized according to Synthesis Examples are shown in Table 1.

(353) TABLE-US-00001 TABLE 1 Compound MS/FAB No. ¹H NMR (CDCl₃, 400 MHz) found calc. BD2 6.57 (d, 2H), 6.78 (d, 2H), 7.13-7.32 1299.11 1299.23 (m, 16H), 7.35-7.47 (m, 12H) BD9 1.35 (s, 6H), 1.39 (s, 6H), 6.56 1355.21 1355.33 (d, 2H), 6.79 (d, 2H), 7.13-7.34 (m, 12H), 7.35-7.48 (m, 12H) BD17 6.57 (d, 2H), 6.78 (d, 2H), 7.13-7.32 1299.12 1299.17 (m, 16H), 7.33-7.42 (m, 10H) BD25 6.58 (d, 2H), 6.79 (d, 2H), 7.13-7.33 1299.05 1299.17 (m, 16H), 7.33-7.42 (m, 10H) BD45 1.35 (s, 12H), 1.39 (s, 12H), 6.58 1411.40 1411.44 (d, 2H), 6.81 (d, 2H), 7.12-7.34 (m, 12H), 7.35-7.49 (m, 12H)

Example 1

(354) As an anode, a glass substrate with 15 Ωcm.² (1,200 Å) ITO thereon, which was manufactured by Corning Inc., was cut to a size of 50 mm×50 mm×0.7 mm, and the glass substrate was sonicated by using isopropyl alcohol and pure water for 5 minutes each, and ultraviolet (UV) light was irradiated for 30 minutes thereto and ozone was exposed thereto for cleaning. The resultant glass substrate was loaded onto a vacuum deposition apparatus.

(355) 2-TNATA, which is a compound in the art, was vacuum-deposited on the substrate to form a hole injection layer having a thickness of 600 Å, and 4,4'-bis[N-(1-naphthyl)-N-phenylamino]biphenyl (hereinafter, referred to as NPB) as a hole transport compound was vacuum-deposited thereon to form a hole transport layer having a thickness of 300 Å.

(356) mCBP as a host and Compound BD2 as a dopant in an amount of 10 wt % were co-deposited

on the hole transport layer to form an emission layer having a thickness of 300 Å.
(357) TSPO1 was vacuum-deposited on the emission layer to form a hole blocking layer having a thickness of 50 Å. Alq.sub.3 was deposited on the hole blocking layer to form an electron transport layer having a thickness of 300 Å, LiF, which is a halogenated alkali metal, was deposited on the electron transport layer to form an electron injection layer having a thickness of 10 Å, and A1 was vacuum-deposited on the electron injection layer to form a LiF/Al cathode electrode having a thickness of 3,000 Å, thereby completing the manufacture of a light-emitting device.

(358) ##STR00157##

Examples 2 to 5 and Comparative Examples 1 and 2

(359) Light-emitting devices were manufactured in the same manner as in Example 1, except that in forming an emission layer, for use as a dopant, corresponding compounds shown in Table 2 were used instead of Compound BD2.

(360) ##STR00158## ##STR00159## ##STR00160##

Evaluation Example

(361) The driving voltage, current density, luminance, efficiency, emission color, and emission wavelength of the light-emitting devices manufactured according to Examples 1 to 5 and Comparative Examples 1 to 3 were measured by using Keithley SMU 236 and a luminance meter PR650, and results thereof are shown in Table 2.

(362) TABLE-US-00002 TABLE 2 Dopant in Driving Current Emission emission voltage density Luminance Efficiency Emission wavelength layer (V) (mA/cm.sup.2) (cd/m.sup.2) (cd/A) color (nm) Example 1 BD2 4.76 50 4032 8.32 Blue 457 Example 2 BD9 4.80 50 4019 8.85 Blue 458 Example 3 BD17 4.97 50 4011 9.12 Bluish 463 green Example 4 BD25 4.95 50 4005 9.01 Bluish 465 green Example 5 BD45 4.90 50 3990 8.26 Bluish 462 green Comparative CE1 5.22 50 3910 7.93 Bluish 464 Example 1 green Comparative CE2 5.29 50 3950 7.75 Bluish 467 Example 2 green

(363) Referring to Table 2, it was found that the light-emitting devices according to an embodiment have a lower driving voltage and higher luminance and efficiency than the light-emitting devices of Comparative Examples.

(364) A light-emitting device including the organometallic compound according to an embodiment may have a low driving voltage, high luminance and efficiency, and a long lifespan.

(365) Embodiments have been disclosed herein, and although terms are employed, they are used and are to be interpreted in a generic and descriptive sense only and not for purpose of limitation. In some instances, as would be apparent by one of ordinary skill in the art, features, characteristics, and/or elements described in connection with an embodiment may be used singly or in combination with features, characteristics, and/or elements described in connection with other embodiments unless otherwise specifically indicated. Accordingly, it will be understood by those of ordinary skill in the art that various changes in form and details may be made without departing from the spirit and scope of the disclosure as set forth in the claims.

Claims

1. An organometallic compound represented by Formula 1: ##STR00161## wherein in Formula 1, M.sub.1 and M.sub.2 are each independently platinum (Pt), palladium (Pd), copper (Cu), silver (Ag), gold (Au), rhodium (Rh), iridium (Ir), ruthenium (Ru), osmium (Os), titanium (Ti), zirconium (Zr), hafnium (Hf), europium (Eu), terbium (Tb), or thulium (Tm), A1 to A4 and B1 to B4 are each independently a C.sub.5-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, Y.sub.10 to Y.sub.12, Y.sub.20 to Y.sub.22, Y.sub.30 to Y.sub.32, Y.sub.40 to Y.sub.42, Y.sub.50 to Y.sub.52, Y.sub.60 to Y.sub.62, Y.sub.70 to Y.sub.72, and Y.sub.80 to Y.sub.82 are each independently C or N, T.sub.1 and T.sub.2 are each independently a single bond, *—O—*, *—S—*, *—C(R.sub.1)(R.sub.2)—*, *—C(R.sub.1)=*, *=C(R.sub.1)—*, *

$C(R.sub.1)=C(R.sub.2)-*$, $*-C(=O)-*$, $*-C(=S)-*$, $*-C\equiv C-*$, $*-B(R.sub.1)-*$, $*-N(R.sub.1)-*$, $*-P(R.sub.1)-*$, $*-Si(R.sub.1)(R.sub.2)-*$, or $*-Ge(R.sub.1)(R.sub.2)-*$, L.sub.10 is a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, a₁₀ is 1, 2, 3, 4, or 5, R.sub.1, R.sub.2, R.sub.10, R.sub.20, R.sub.30, R.sub.40, R.sub.50, R.sub.60, R.sub.70, and R.sub.80 are each independently hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.20 alkoxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.10 cycloalkyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.10 heterocycloalkyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.10 cycloalkenyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.10 heterocycloalkenyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 aryl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 aryloxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 arylthio group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heteroaryl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heteroaryloxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heteroarylthio group unsubstituted or substituted with at least one R.sub.10a, a monovalent non-aromatic condensed polycyclic group unsubstituted or substituted with at least one R.sub.10a, a monovalent non-aromatic condensed heteropolycyclic group unsubstituted or substituted with at least one R.sub.10a, —Si(Q.sub.41)(Q.sub.42)(Q.sub.43), —N(Q.sub.41)(Q.sub.42), —B(Q.sub.41)(Q.sub.42), —C(=O)(Q.sub.41), —S(=O).sub.2(Q.sub.41), or —P(=O)(Q.sub.41)(Q.sub.42), two neighboring substituents among R.sub.1, R.sub.2, R.sub.10, R.sub.20, R.sub.30, R.sub.40, R.sub.50, R.sub.60, R.sub.70, and R.sub.80 are optionally linked to each other to form a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, b₁₀, b₂₀, b₃₀, b₄₀, b₅₀, b₆₀, b₇₀, and b₈₀ are each independently 1, 2, 3, 4, 5, 6, 7, or 8, * and *' each indicate a binding site to a neighboring atom, and R.sub.10a is: deuterium (-D), —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, or a nitro group; a C.sub.1-C.sub.6 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, or a C.sub.1-C.sub.60 alkoxy group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryloxy group, a C.sub.1-C.sub.60 heteroarylthio group, —Si(Q.sub.11)(Q.sub.12)(Q.sub.13), —N(Q.sub.11)(Q.sub.12), —B(Q.sub.11)(Q.sub.12), —C(=O)(Q.sub.11), —S(=O).sub.2(Q.sub.11), —P(=O)(Q.sub.11)(Q.sub.12), or a combination thereof; a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryloxy group, or a C.sub.1-C.sub.60 heteroarylthio group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.1-C.sub.60 heteroaryloxy group, a C.sub.1-C.sub.60 heteroarylthio group, —Si(Q.sub.21)(Q.sub.22)(Q.sub.23), —N(Q.sub.21)(Q.sub.22), —B(Q.sub.21)(Q.sub.22), —C(=O)(Q.sub.21), —S(=O).sub.2(Q.sub.21), —P(=O)(Q.sub.21)(Q.sub.22), or a combination thereof; or —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), or —P(=O)(Q.sub.31)(Q.sub.32), wherein Q.sub.11 to Q.sub.13, Q.sub.21 to Q.sub.23, Q.sub.31 to Q.sub.33, and Q.sub.41 to Q.sub.43 are each independently: hydrogen; deuterium; —F; —Cl; —

- Br; —I; a hydroxyl group; a cyano group; a nitro group; a C.sub.1-C.sub.60 alkyl group; a C.sub.2-C.sub.60 alkenyl group; a C.sub.2-C.sub.60 alkynyl group; a C.sub.1-C.sub.60 alkoxy group; or a C.sub.3-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, each unsubstituted or substituted with deuterium, —F, a cyano group, a C.sub.1-C.sub.60 alkyl group, a C.sub.1-C.sub.60 alkoxy group, a phenyl group, a biphenyl group, or a combination thereof.
2. The organometallic compound of claim 1, wherein M.sub.1 and M.sub.2 are each independently Pt or Pd.
3. The organometallic compound of claim 1, wherein A1 to A4 are each independently: a 5-membered heterocyclic group, or a polycyclic C.sub.1-C.sub.30 heterocyclic group comprising a 5-membered heterocyclic group.
4. The organometallic compound of claim 1, wherein A1 and A4 are each independently a group represented by one of Formulae 3-1 to 3-3, and A2 and A3 are each independently a group represented by one of Formulae 3-4 and 3-5: ##STR00162## wherein in Formulae 3-1 to 3-5, X.sub.31 to X.sub.39 are each independently C(Z.sub.32) or N, Z.sub.31 and Z.sub.32 are each independently the same as described in connection with R.sub.10 in Formula 1, and *, *' and *'' each indicate a binding site to a neighboring atom.
5. The organometallic compound of claim 1, wherein a bond between M.sub.1 and Y.sub.10 is a coordinate bond, a bond between M.sub.1 and Y.sub.20 is a coordinate bond, a bond between M.sub.2 and Y.sub.30 is a coordinate bond, a bond between M.sub.2 and Y.sub.40 is a coordinate bond, a bond between M.sub.1 and Y.sub.50 is a covalent bond, a bond between M.sub.1 and Y.sub.60 is a covalent bond, a bond between M.sub.2 and Y.sub.70 is a covalent bond, and a bond between M.sub.2 and Y.sub.80 is a covalent bond.
6. The organometallic compound of claim 1, wherein T.sub.1 and T.sub.2 are each independently *—O—*' or *—S—*'.
7. The organometallic compound of claim 1, wherein L.sub.10 is a C.sub.1-C.sub.20alkylene group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.20 alkenylene group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.20 alkynylene group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.10 cycloalkylene group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.10 heterocycloalkylene group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.10 cycloalkenylene group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.10 heterocycloalkenylene group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 arylene group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heteroarylene group unsubstituted or substituted with at least one R.sub.10a, a divalent non-aromatic condensed polycyclic group unsubstituted or substituted with at least one R.sub.10a, or a divalent non-aromatic condensed heteropolycyclic group unsubstituted or substituted with at least one R.sub.10a.
8. The organometallic compound of claim 1, wherein L.sub.10 is a group represented by one of Formulae 4-1 to 4-3: ##STR00163## ##STR00164## wherein in Formulae 4-1 to 4-3, Y.sub.91 is C(R.sub.91) or N, Y.sub.92 is C(R.sub.92) or N, Y.sub.93 is C(R.sub.93) or N, Y.sub.94 is C(R.sub.94) or N, R.sub.91 to R.sub.94 are each independently the same as described in connection with R.sub.10a in Formula 1, and * and *' each indicate a binding site to a neighboring atom.
9. The organometallic compound of claim 1, wherein R.sub.1, R.sub.2, R.sub.10, R.sub.20, R.sub.30, R.sub.40, R.sub.50, R.sub.60, R.sub.70, and R.sub.80 are each independently: hydrogen, deuterium, —F, —Cl, —Br, —I, a cyano group, a C.sub.1-C.sub.20 alkyl group, or a C.sub.1-C.sub.20 alkoxy group; or a C.sub.1-C.sub.20 alkyl group or a C.sub.1-C.sub.20 alkoxy group, each substituted with deuterium, —F, —Cl, —Br, —I, a cyano group, a phenyl group, a biphenyl group, or a combination thereof; or a group represented by one of Formulae 5-1 to 5-26 and 6-1 to 6-55, and wherein neighboring two or more of R.sub.1, R.sub.2, R.sub.10, R.sub.20, R.sub.30, R.sub.40, R.sub.50, R.sub.60, R.sub.70, and R.sub.80 are optionally linked to each other to form a

cyclopentane group, a cyclohexane group, a benzene group, a naphthylene group, a fluorene group, a pyridine group, a pyrimidine group, a triazine group, or a carbazole group: ##STR00165##
##STR00166## ##STR00167## ##STR00168## ##STR00169## ##STR00170## ##STR00171##
##STR00172## ##STR00173## ##STR00174## wherein in Formulae 5-1 to 5-26 and 6-1 to 6-55, X.sub.51 and X.sub.52 are each independently O, S, C(Z.sub.53)(Z.sub.54), N(Z.sub.53), or Si(Z.sub.53)(Z.sub.54), Z.sub.51 to Z.sub.54 are each independently hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C.sub.1-C.sub.20 alkyl group, a C.sub.2-C.sub.20 alkenyl group, a C.sub.2-C.sub.20 alkynyl group, a C.sub.1-C.sub.20 alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a phenanthrenyl group, an anthracenyl group, a triphenylenyl group, a pyridinyl group, a pyrimidinyl group, a carbazolyl group, or a triazinyl group, e2 is 1 or 2, e3 is 1, 2, or 3, e4 is 1, 2, 3, or 4, e5 is 1, 2, 3, 4, or 5, e6 is 1, 2, 3, 4, 5, or 6, e7 is 1, 2, 3, 4, 5, 6, or 7, e9 is 1, 2, 3, 4, 5, 6, 7, 8, or 9, and * indicates a binding site to a neighboring atom.

10. The organometallic compound of claim 1, wherein the organometallic compound is represented by one of Formulae 11-1 to 11-3: ##STR00175## ##STR00176## ##STR00177## wherein in Formulae 11-1 to 11-3, Y.sub.23 is C(R.sub.23) or N, Y.sub.24 is C(R.sub.24) or N, Y.sub.25 is C(R.sub.25) or N, Y.sub.26 is C(R.sub.26) or N, Y.sub.27 is C(R.sub.27) or N, Y.sub.28 is C(R.sub.28) or N, Y.sub.33 is C(R.sub.33) or N, Y.sub.34 is C(R.sub.34) or N, Y.sub.35 is C(R.sub.35) or N, Y.sub.36 is C(R.sub.36) or N, Y.sub.37 is C(R.sub.37) or N, Y.sub.38 is C(R.sub.38) or N, R.sub.23 to R.sub.28 are each independently the same as described in connection with R.sub.20 in Formula 1, and R.sub.33 to R.sub.38 are each independently the same as described in connection with R.sub.30 in Formula 1.

11. The organometallic compound of claim 1, wherein the organometallic compound is electrically neutral.

12. The organometallic compound of claim 1, wherein the organometallic compound is one selected from Compounds BD1 to BD104: ##STR00178## ##STR00179## ##STR00180##
##STR00181## ##STR00182## ##STR00183## ##STR00184## ##STR00185## ##STR00186##
##STR00187## ##STR00188## ##STR00189## ##STR00190## ##STR00191## ##STR00192##
##STR00193## ##STR00194## ##STR00195## ##STR00196## ##STR00197## ##STR00198##
##STR00199## ##STR00200## ##STR00201## ##STR00202## ##STR00203## ##STR00204##
##STR00205## ##STR00206## ##STR00207## ##STR00208## ##STR00209## ##STR00210##
##STR00211## ##STR00212## ##STR00213## ##STR00214## ##STR00215## ##STR00216##
##STR00217## ##STR00218## ##STR00219## ##STR00220## ##STR00221##

13. A light-emitting device comprising: a first electrode; a second electrode facing the first electrode; an interlayer disposed between the first electrode and the second electrode and comprising an emission layer; and the organometallic compound of claim 1.

14. The light-emitting device of claim 13, wherein the first electrode is an anode, the second electrode is a cathode, and the interlayer further comprises: a hole transport region disposed between the first electrode and the emission layer and comprising a hole injection layer, a hole transport layer, an emission auxiliary layer, an electron blocking layer, or a combination thereof; and an electron transport region disposed between the emission layer and the second electrode and comprising a hole blocking layer, an electron transport layer, an electron injection layer, or a combination thereof.

15. The light-emitting device of claim 13, wherein the emission layer comprises the organometallic compound.

16. The light-emitting device of claim 15, wherein the emission layer comprises a host and a dopant, and the dopant comprises the organometallic compound.

17. The light-emitting device of claim 14, wherein the hole transport region further comprises a p-dopant having a lowest unoccupied molecular orbital (LUMO) energy level equal to or less than

about -3.5 eV.

18. The light-emitting device of claim 14, wherein the electron transport region comprises the electron transport layer and the electron injection layer, and at least one layer of the electron transport layer and the electron injection layer comprises an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal compound, an alkaline earth-metal compound, a rare earth metal compound, an alkali metal complex, an alkaline earth-metal complex, a rare earth metal complex, or a combination thereof.

19. An electronic apparatus comprising the light-emitting device of claim 13.

20. The electronic apparatus of claim 19, further comprising a thin-film transistor, wherein the thin-film transistor comprises a source electrode and a drain electrode, and the first electrode of the light-emitting device is electrically connected to the source electrode or the drain electrode.
